

OCT ANALYSER — DUAL-HEAD POPULATION TELEMETRY DASHBOARD

Hierarchical Multi-Task ConvNeXt Evaluation & Explainability Analysis

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EXECUTIVE METRIC SUMMARY (VECTORIZED VALIDATION ON 17,761 B-SCANS)

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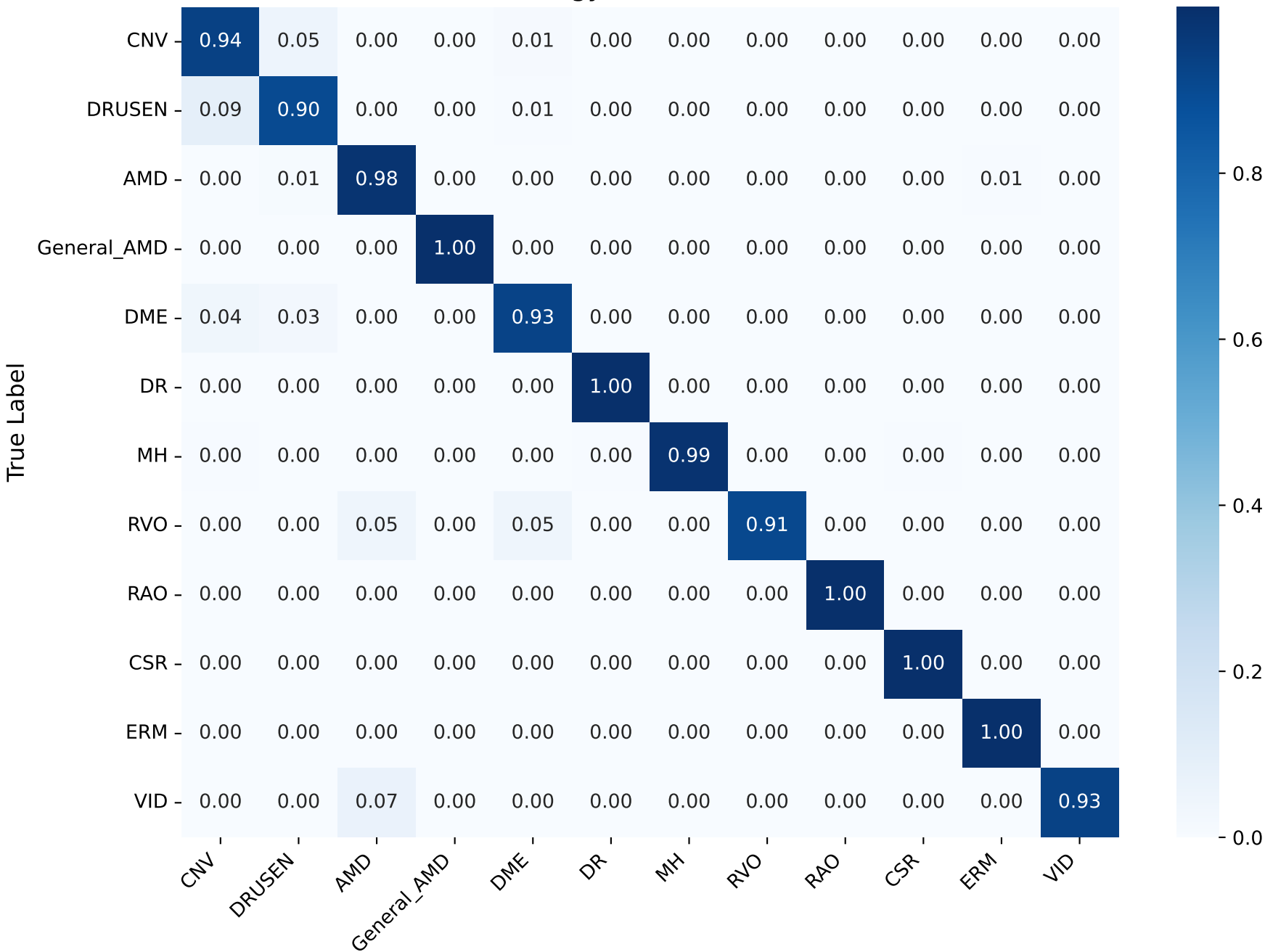
- H1 Gatekeeper Triage Accuracy : 96.60% | H1 Macro-F1 : 0.9609
- H2 Pathology Multi-Class Accuracy: 93.30% | H2 Macro-F1 : 0.9489
- Active Pathology Classes : 12 / 12 (100% active, 0 dead classes)

PER-CLASS PERFORMANCE BREAKDOWN:

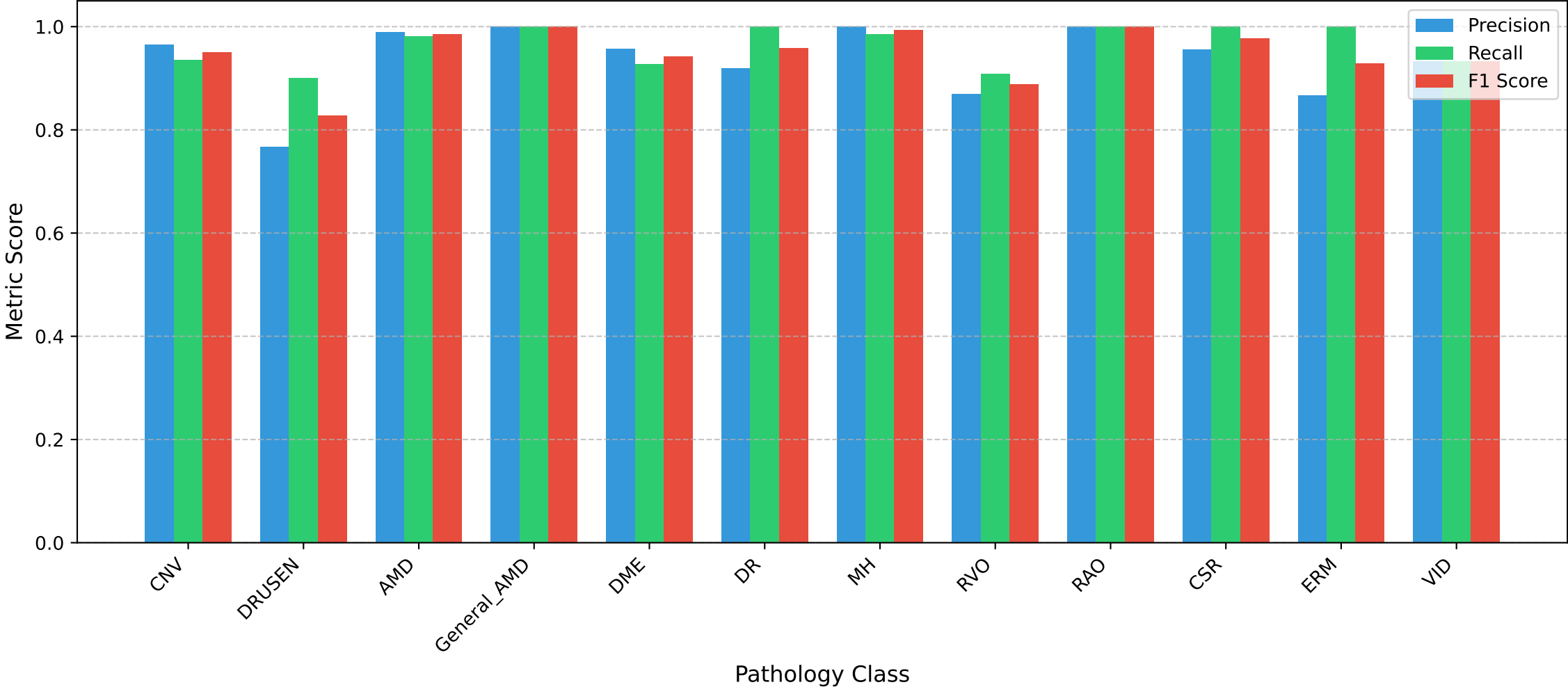
- CNV	F1: 0.9505	Prec: 0.9652	Rec: 0.9363	AUC: 0.9880	Support: 7314
- DRUSEN	F1: 0.8284	Prec: 0.7669	Rec: 0.9007	AUC: 0.9861	Support: 1651
- AMD	F1: 0.9861	Prec: 0.9900	Rec: 0.9822	AUC: 0.9997	Support: 505
- General_AMD	F1: 1.0000	Prec: 1.0000	Rec: 1.0000	AUC: 1.0000	Support: 9
- DME	F1: 0.9422	Prec: 0.9569	Rec: 0.9280	AUC: 0.9958	Support: 2319
- DR	F1: 0.9583	Prec: 0.9200	Rec: 1.0000	AUC: 1.0000	Support: 23
- MH	F1: 0.9930	Prec: 1.0000	Rec: 0.9862	AUC: 1.0000	Support: 217
- RVO	F1: 0.8889	Prec: 0.8696	Rec: 0.9091	AUC: 0.9998	Support: 22
- RAO	F1: 1.0000	Prec: 1.0000	Rec: 1.0000	AUC: 1.0000	Support: 3
- CSR	F1: 0.9778	Prec: 0.9565	Rec: 1.0000	AUC: 1.0000	Support: 22
- ERM	F1: 0.9286	Prec: 0.8667	Rec: 1.0000	AUC: 1.0000	Support: 26
- VID	F1: 0.9333	Prec: 0.9333	Rec: 0.9333	AUC: 0.9999	Support: 15

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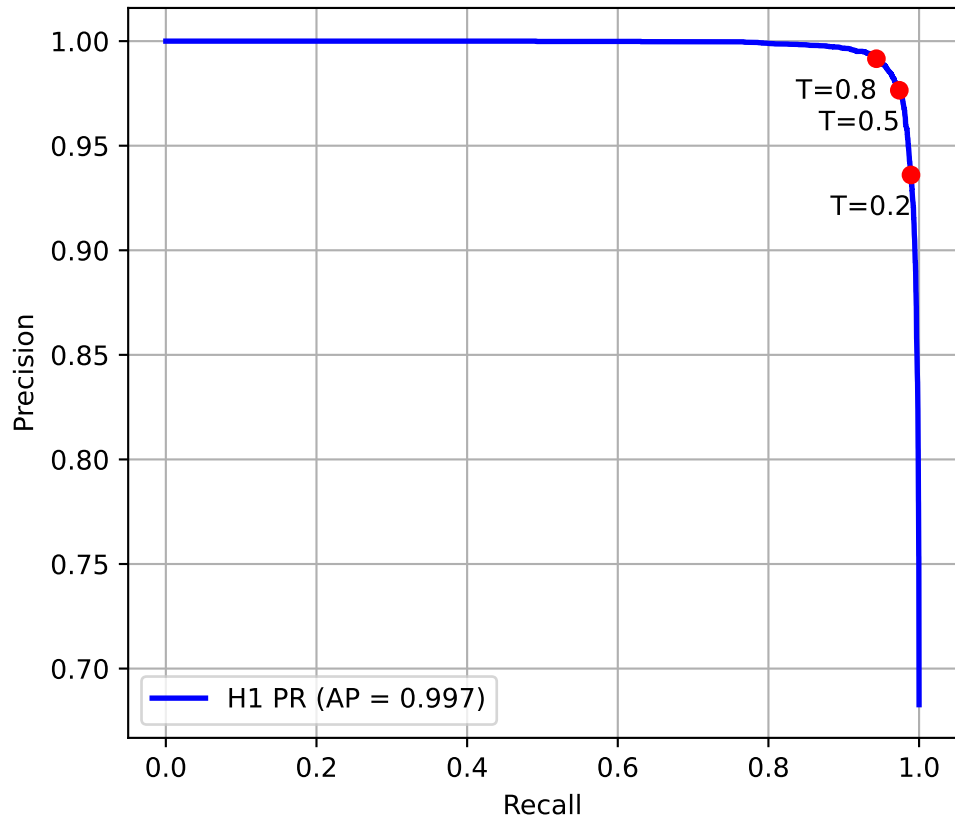
H2 Granular Pathology Normalized Confusion Matrix



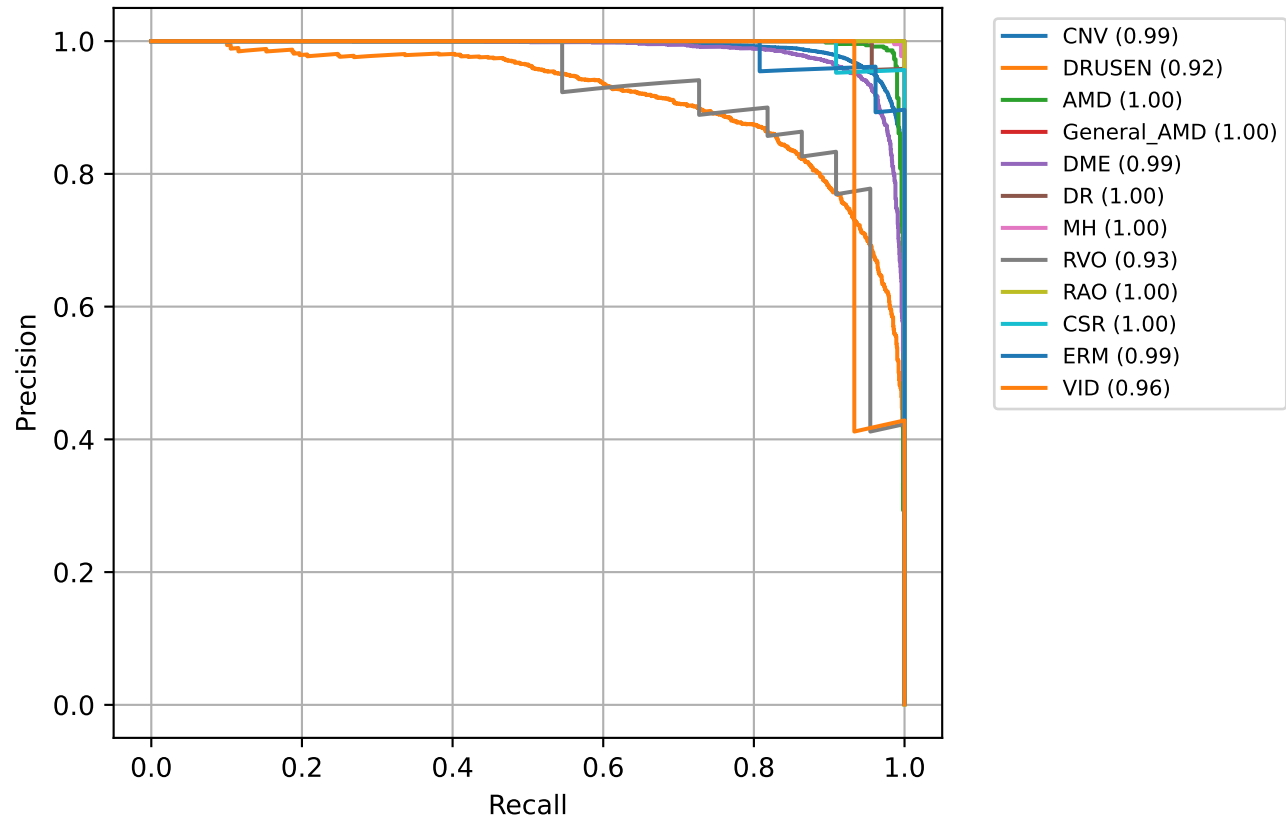
H2 Pathology Per-Class Precision, Recall, and F1 Score



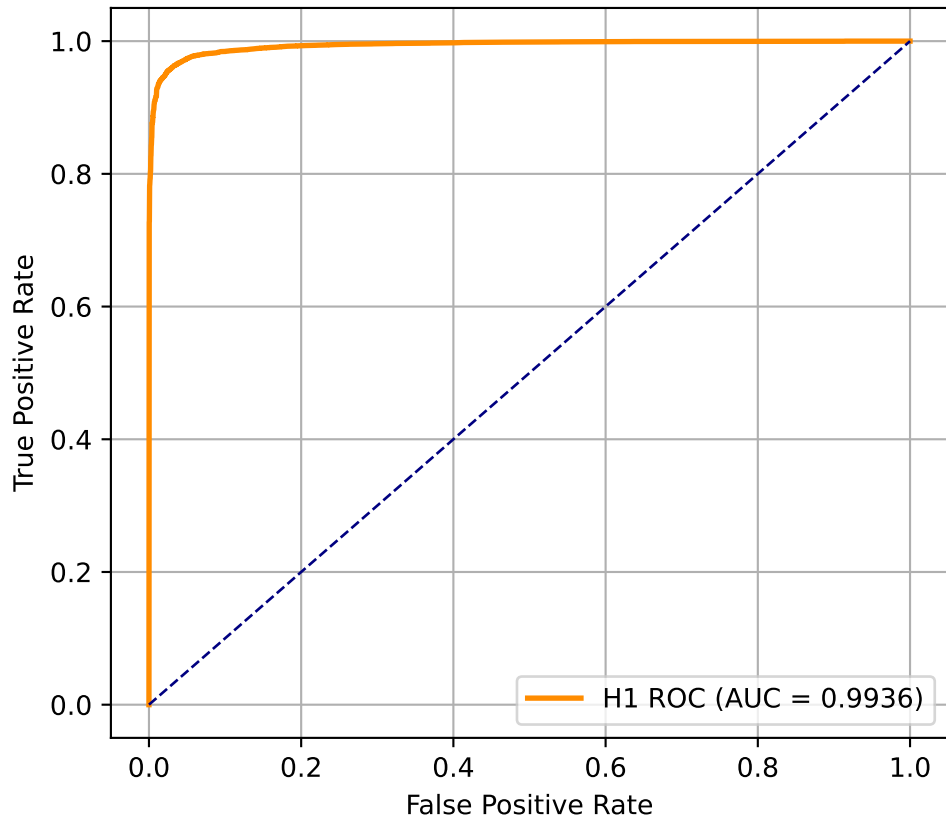
H1 Triage PR Curve



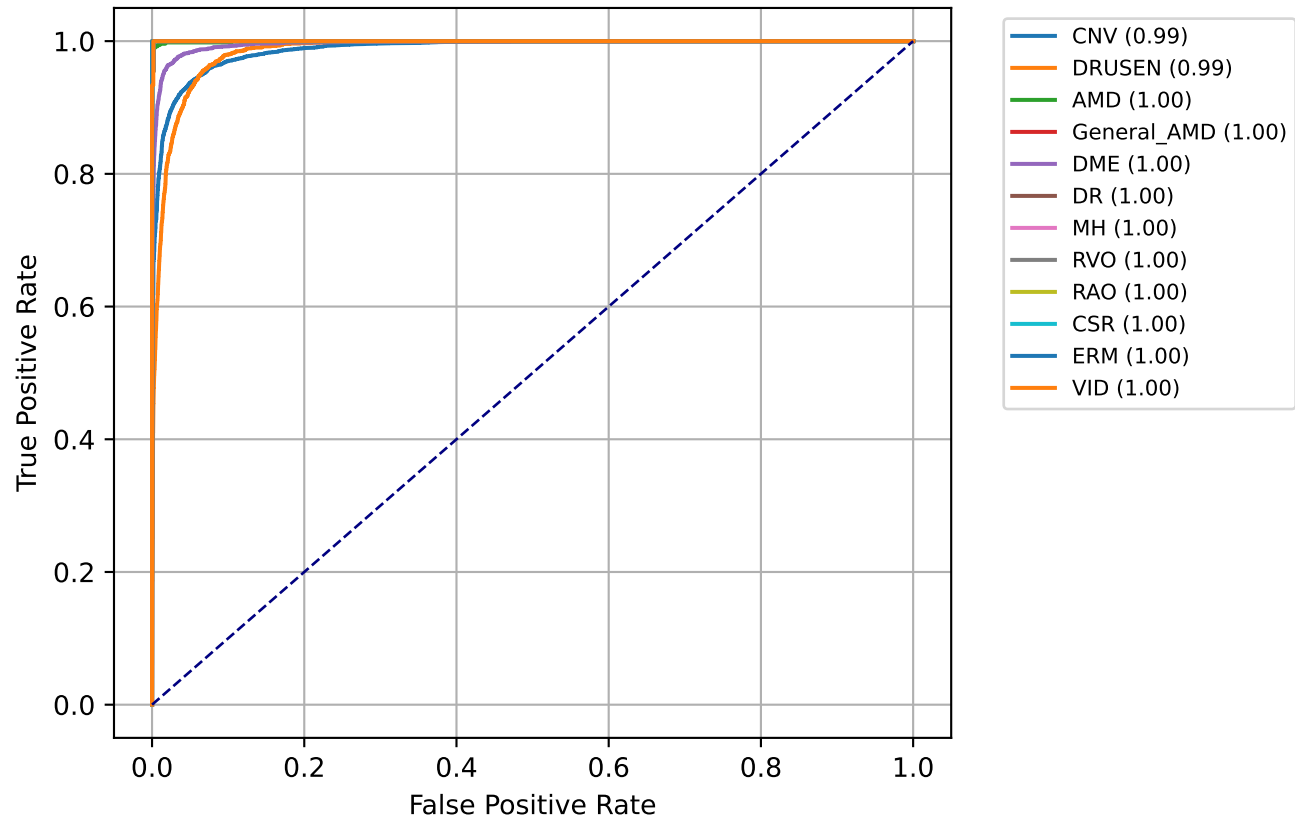
H2 Granular Pathology PR Curves



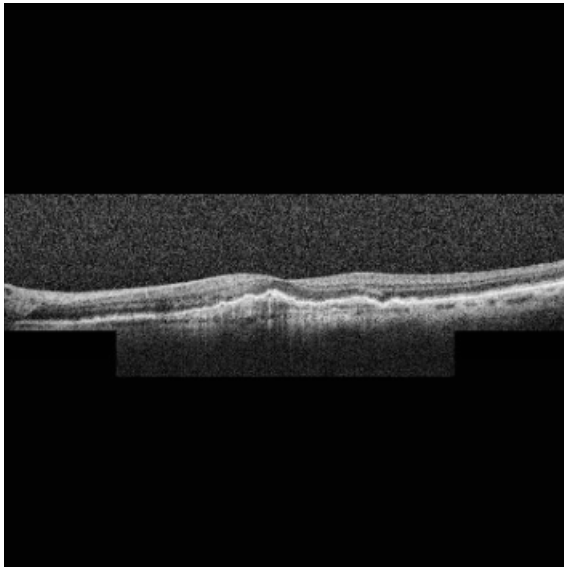
H1 Triage ROC Curve



H2 Granular Pathology ROC Curves



Original OCT Scan
True: CNV



PATIENT CASE STUDY REPORT

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[H1 - Gatekeeper Triage]

Status : Abnormal (100.0%)

Decision : Pass to H2

[H2 - Pathology Routing]

True : CNV

Top-1 : CNV (87.7%)

Top-2 : DRUSEN (10.5%)

Status : [CORRECT]

[CAM Ambiguity Quantification]

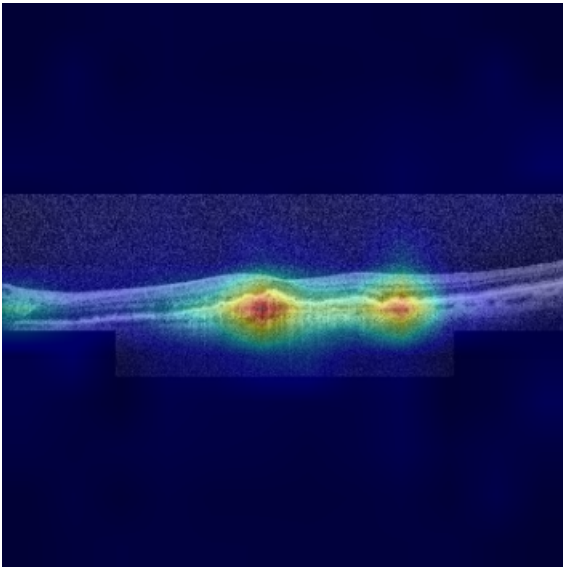
Cosine Sim : 0.503

Overlap IoU : 0.045

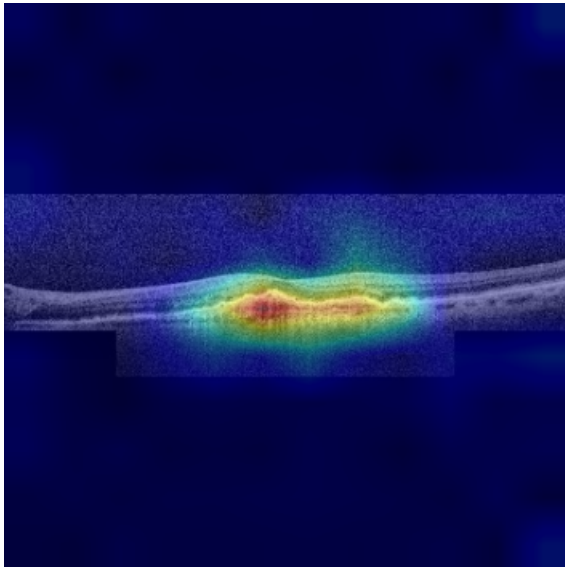
Pathology Probabilities (>5%):

- CNV : 87.7%
- DRUSEN : 10.5%

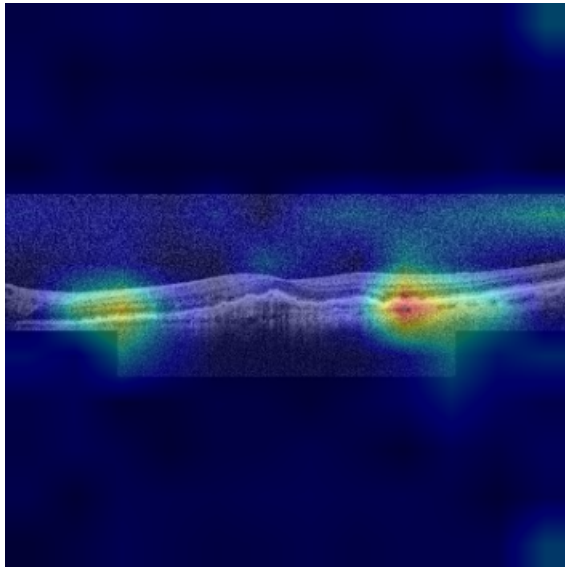
H1 Grad-CAM
(Triage)



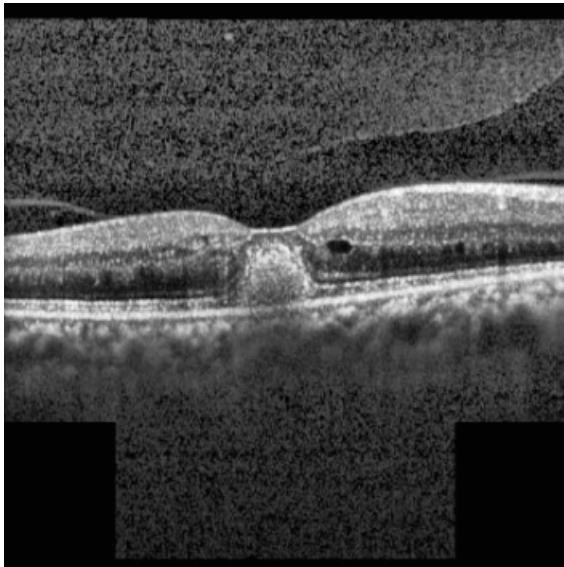
H2 Grad-CAM
CNV (87.7%)



H2 Grad-CAM
DRUSEN (10.5%)



Original OCT Scan
True: CNV



PATIENT CASE STUDY REPORT

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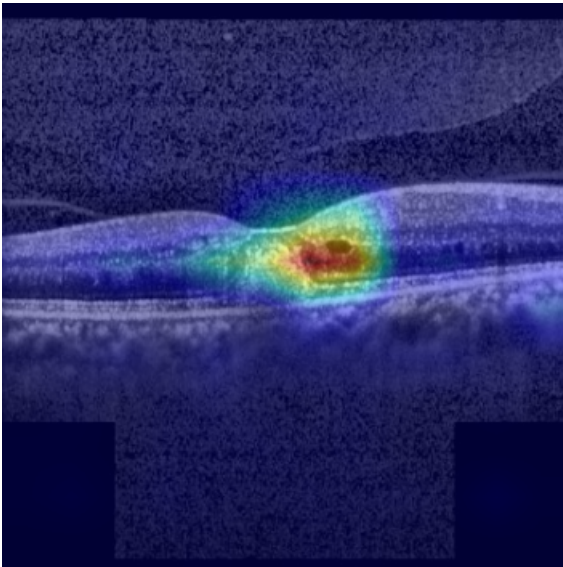
[H1 - Gatekeeper Triage]
Status : Abnormal (100.0%)
Decision : Pass to H2

[H2 - Pathology Routing]
True : CNV
Top-1 : DME (79.8%)
Top-2 : CNV (15.3%)
Status : [MISMATCH]

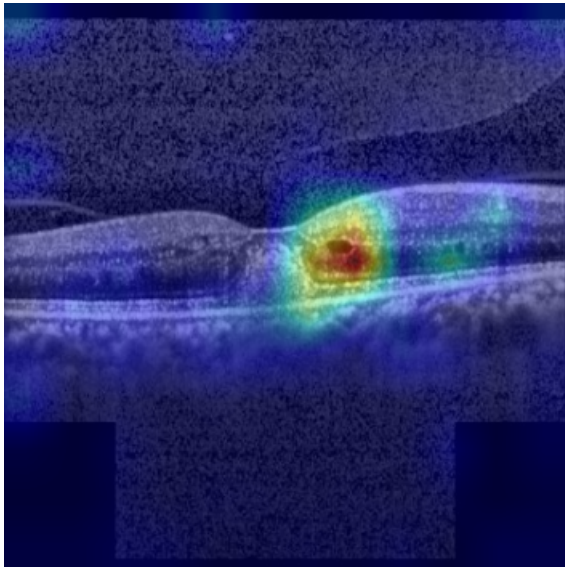
[CAM Ambiguity Quantification]
Cosine Sim : 0.435
Overlap IoU : 0.091

Pathology Probabilities (>5%):
- DME : 79.8%
- CNV : 15.3%

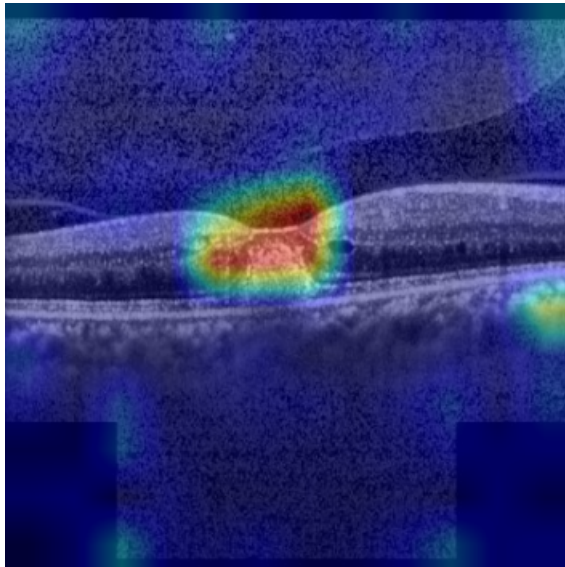
H1 Grad-CAM
(Triage)



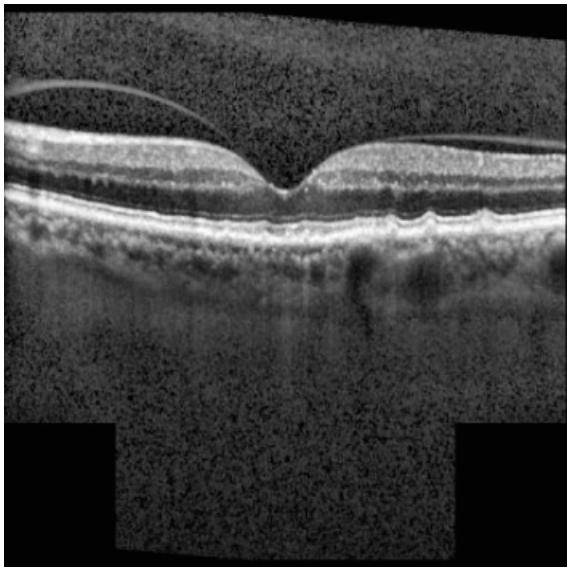
H2 Grad-CAM
DME (79.8%)



H2 Grad-CAM
CNV (15.3%)



Original OCT Scan
True: DRUSEN



PATIENT CASE STUDY REPORT
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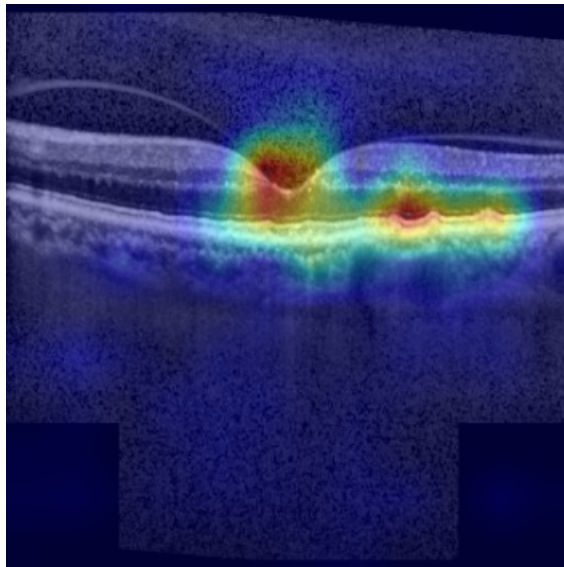
[H1 - Gatekeeper Triage]
Status : Abnormal (99.0%)
Decision : Pass to H2

[H2 - Pathology Routing]
True : DRUSEN
Top-1 : DRUSEN (97.4%)
Top-2 : CNV (1.1%)
Status : [CORRECT]

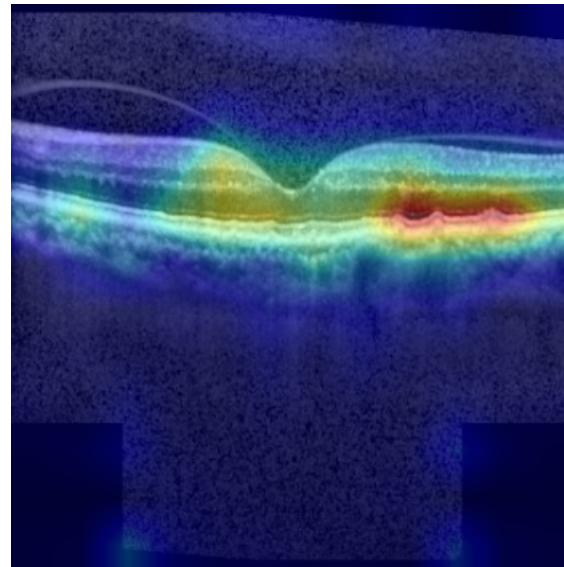
[CAM Ambiguity Quantification]
Cosine Sim : 0.589
Overlap IoU : 0.078

Pathology Probabilities (>5%):
- DRUSEN : 97.4%

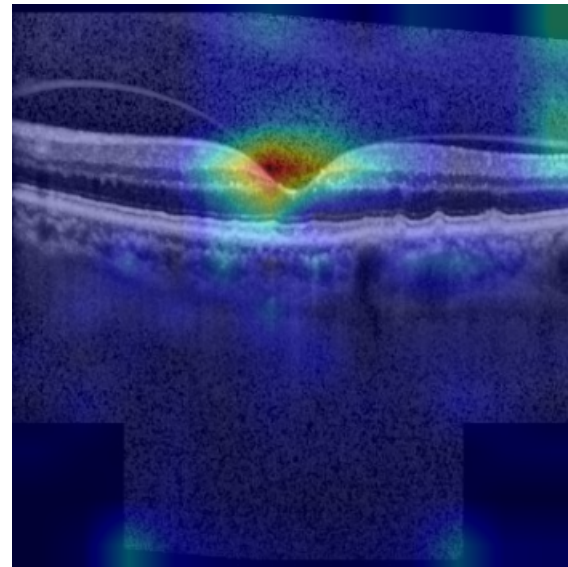
H1 Grad-CAM
(Triage)



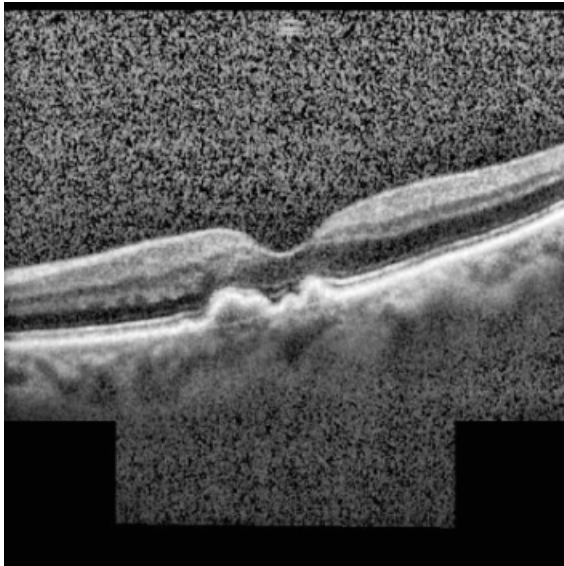
H2 Grad-CAM
DRUSEN (97.4%)



H2 Grad-CAM
CNV (1.1%)



Original OCT Scan
True: DRUSEN



PATIENT CASE STUDY REPORT

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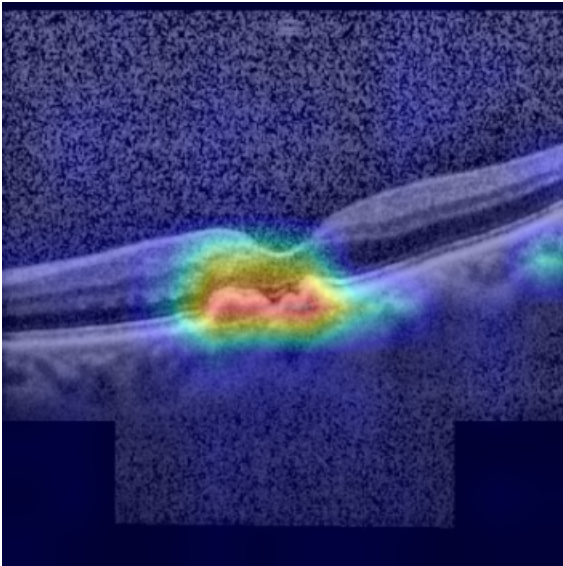
[H1 - Gatekeeper Triage]
Status : Abnormal (99.6%)
Decision : Pass to H2

[H2 - Pathology Routing]
True : DRUSEN
Top-1 : CNV (54.4%)
Top-2 : DRUSEN (42.7%)
Status : [MISMATCH]

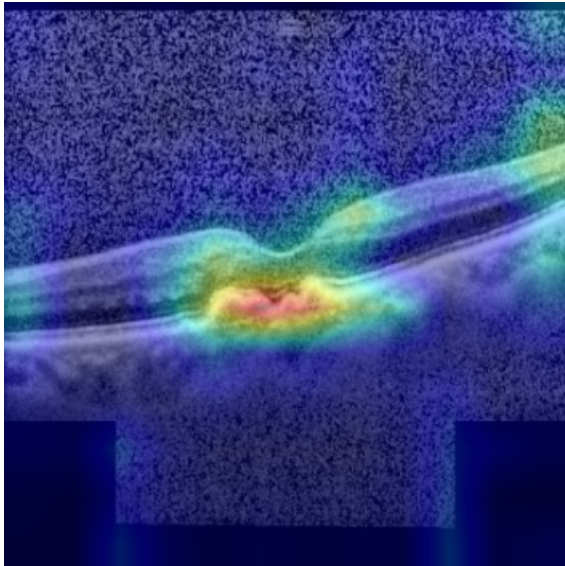
[CAM Ambiguity Quantification]
Cosine Sim : 0.849
Overlap IoU : 0.262

Pathology Probabilities (>5%):
- CNV : 54.4%
- DRUSEN : 42.7%

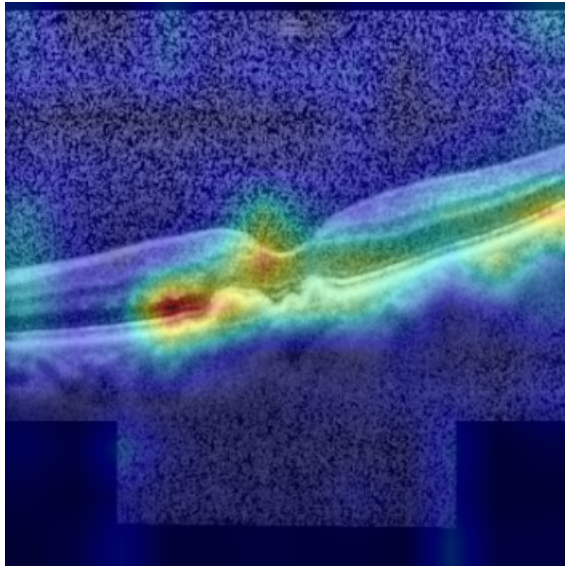
H1 Grad-CAM
(Triage)



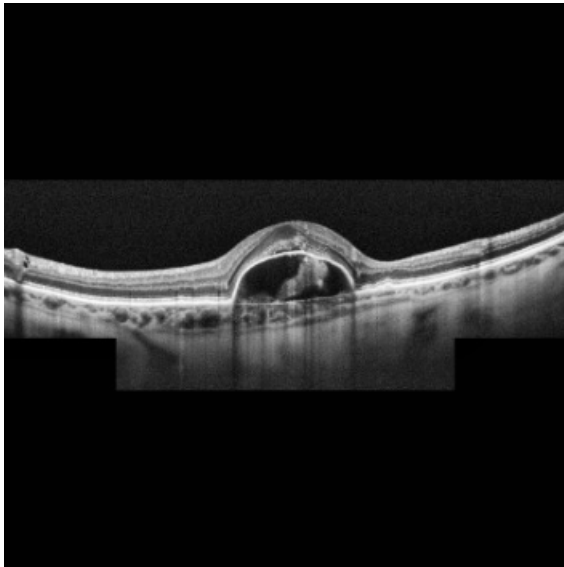
H2 Grad-CAM
CNV (54.4%)



H2 Grad-CAM
DRUSEN (42.7%)



Original OCT Scan
True: AMD



PATIENT CASE STUDY REPORT

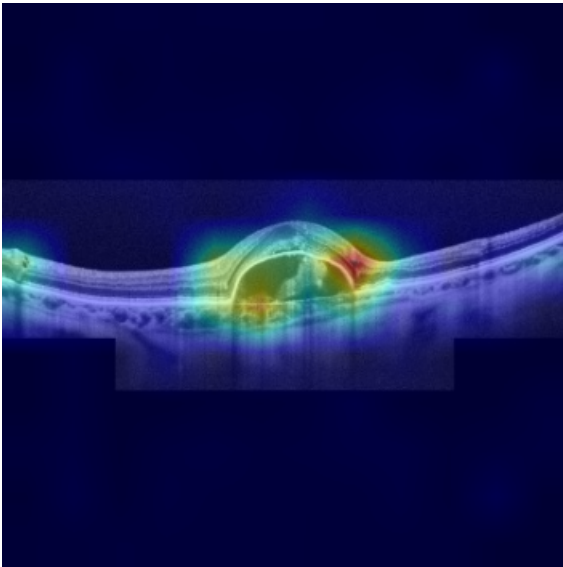
[H1 - Gatekeeper Triage]
Status : Abnormal (100.0%)
Decision : Pass to H2

[H2 - Pathology Routing]
True : AMD
Top-1 : DME (45.0%)
Top-2 : CNV (30.3%)
Status : [MISMATCH]

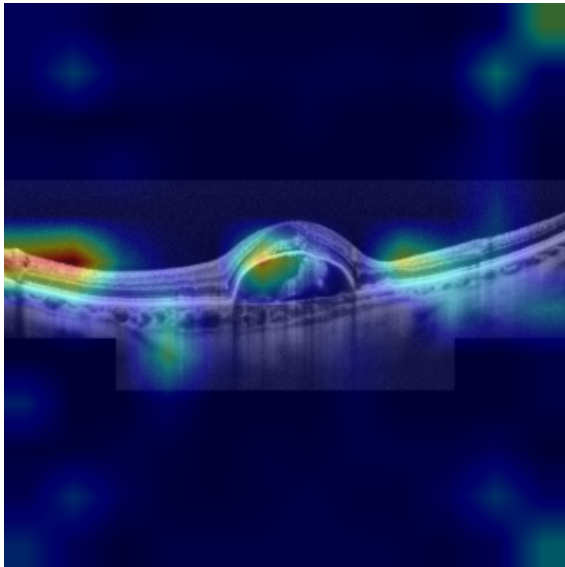
[CAM Ambiguity Quantification]
Cosine Sim : 0.410
Overlap IoU : 0.054

Pathology Probabilities (>5%):
- DME : 45.0%
- CNV : 30.3%
- AMD : 11.8%
- MH : 9.2%

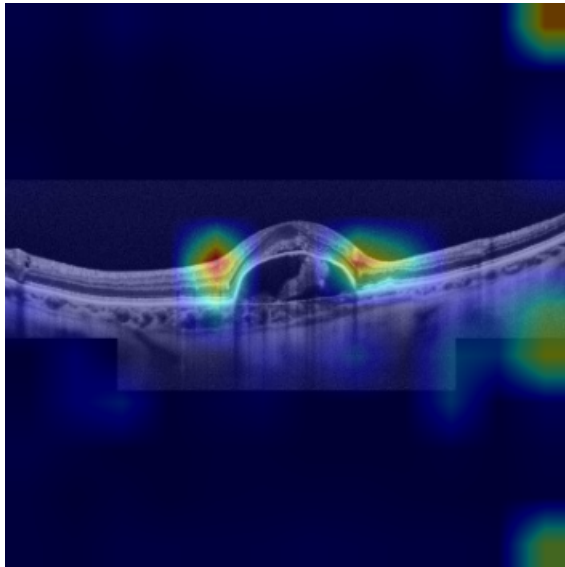
H1 Grad-CAM
(Triage)



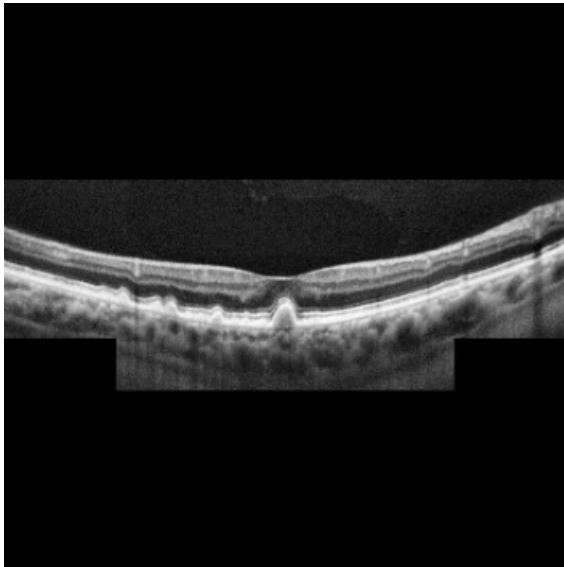
H2 Grad-CAM
DME (45.0%)



H2 Grad-CAM
CNV (30.3%)



Original OCT Scan
True: AMD



PATIENT CASE STUDY REPORT
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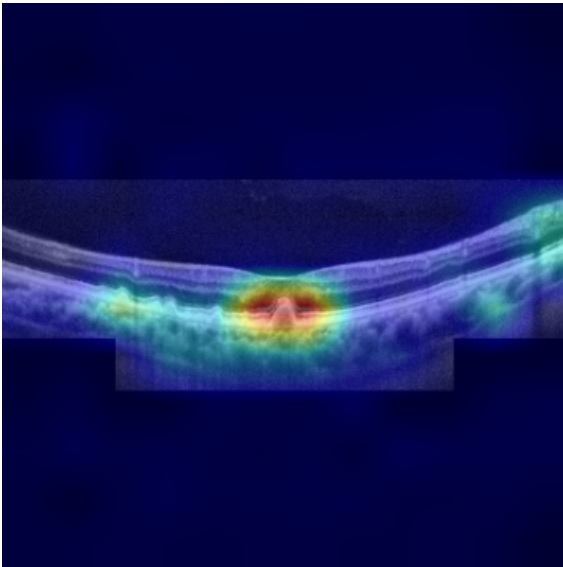
[H1 - Gatekeeper Triage]
Status : Abnormal (100.0%)
Decision : Pass to H2

[H2 - Pathology Routing]
True : AMD
Top-1 : DRUSEN (96.9%)
Top-2 : AMD (1.1%)
Status : [MISMATCH]

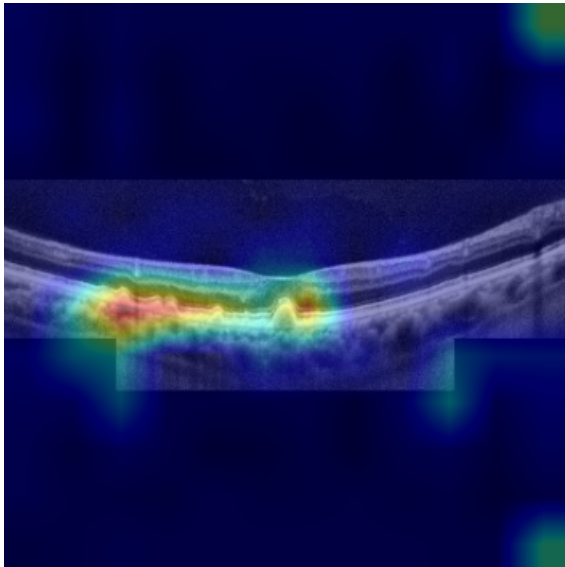
[CAM Ambiguity Quantification]
Cosine Sim : 0.576
Overlap IoU : 0.063

Pathology Probabilities (>5%):
- DRUSEN : 96.9%

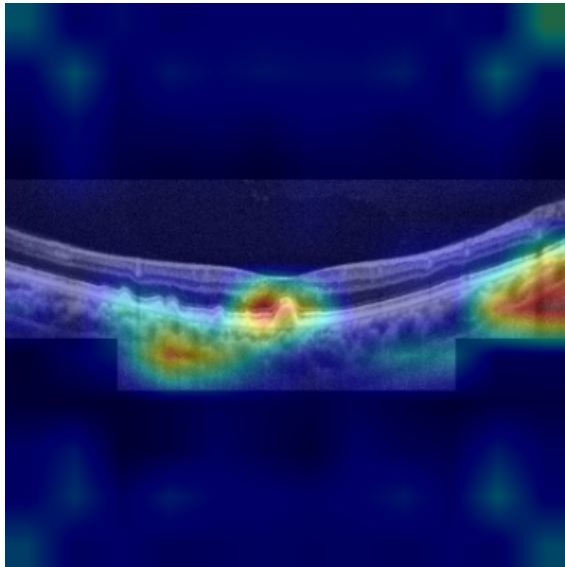
H1 Grad-CAM
(Triage)



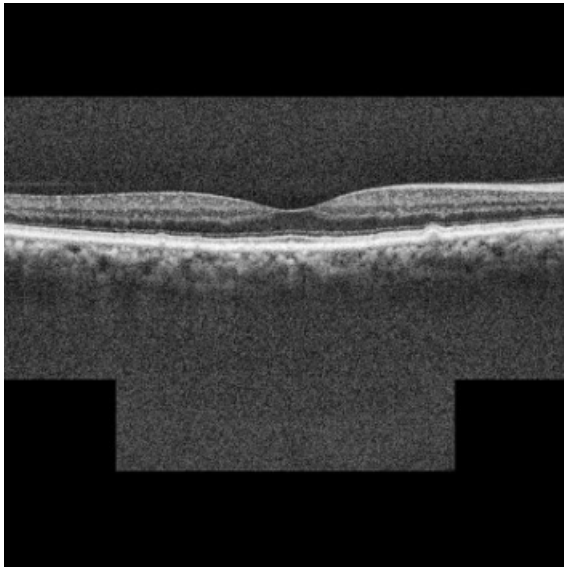
H2 Grad-CAM
DRUSEN (96.9%)



H2 Grad-CAM
AMD (1.1%)

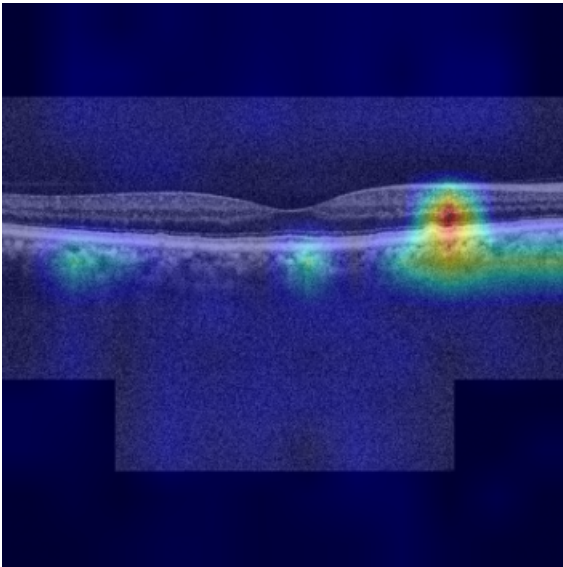


Original OCT Scan
True: General_AMD

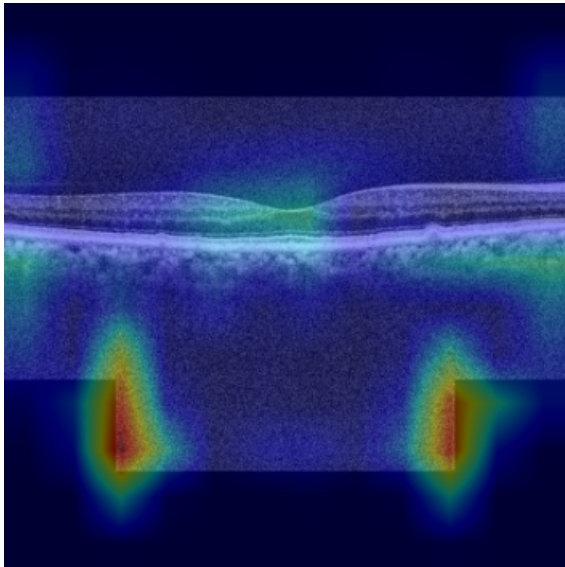


PATIENT CASE STUDY REPORT	
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[H1 - Gatekeeper Triage]	
Status : Abnormal (58.0%)	
Decision : Pass to H2	
[H2 - Pathology Routing]	
True : General_AMD	
Top-1 : General_AMD (67.2%)	
Top-2 : CSR (28.4%)	
Status : [CORRECT]	
[CAM Ambiguity Quantification]	
Cosine Sim : 0.912	
Overlap IoU : 0.543	
Pathology Probabilities (>5%):	
- General_AMD : 67.2%	
- CSR : 28.4%	

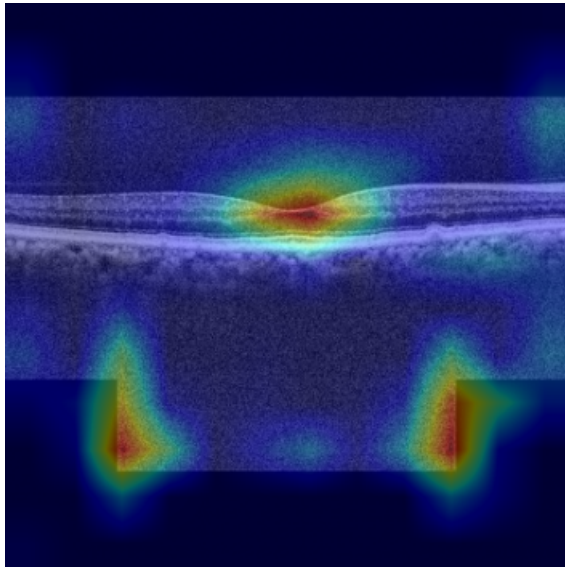
H1 Grad-CAM
(Triage)



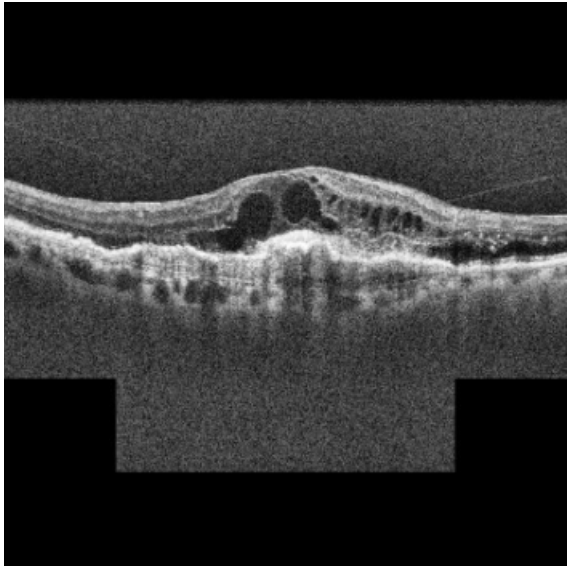
H2 Grad-CAM
General_AMD (67.2%)



H2 Grad-CAM
CSR (28.4%)



Original OCT Scan
True: General_AMD



PATIENT CASE STUDY REPORT
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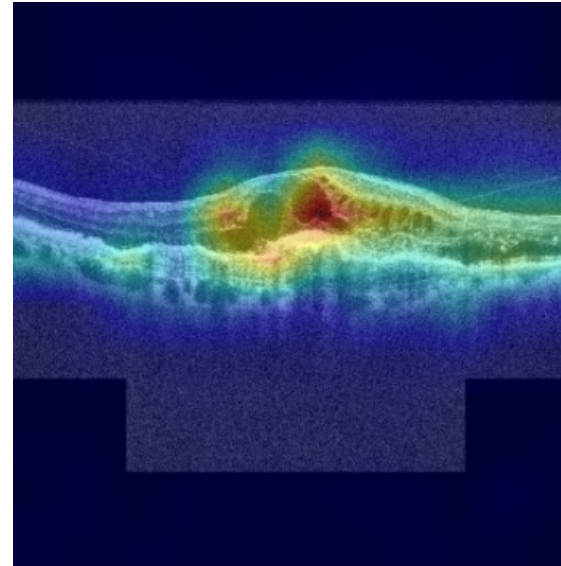
[H1 - Gatekeeper Triage]
Status : Abnormal (100.0%)
Decision : Pass to H2

[H2 - Pathology Routing]
True : General_AMD
Top-1 : General_AMD (95.1%)
Top-2 : DR (3.3%)
Status : [CORRECT]

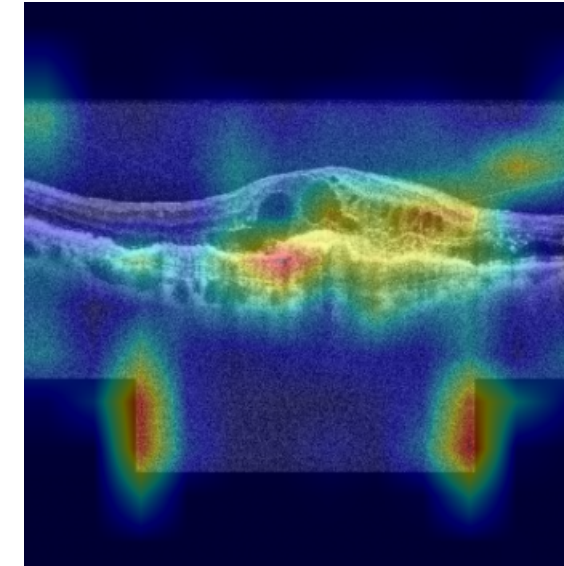
[CAM Ambiguity Quantification]
Cosine Sim : 0.680
Overlap IoU : 0.264

Pathology Probabilities (>5%):
- General_AMD : 95.1%

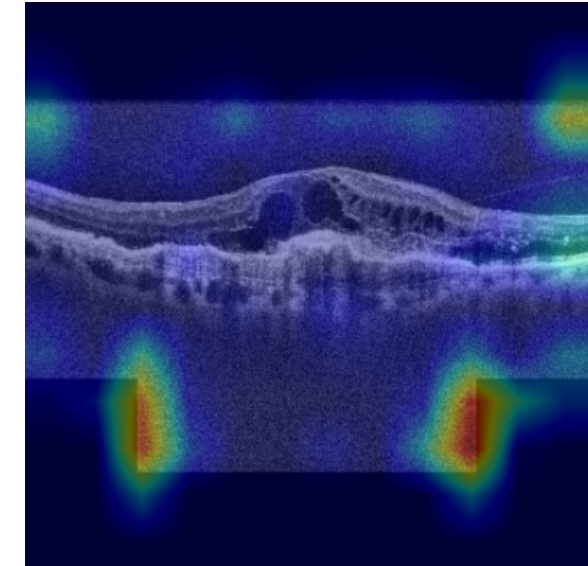
H1 Grad-CAM
(Triage)



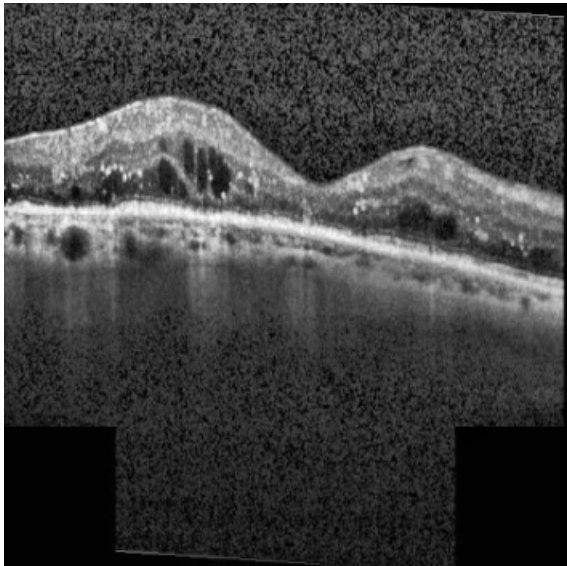
H2 Grad-CAM
General_AMD (95.1%)



H2 Grad-CAM
DR (3.3%)



Original OCT Scan
True: DME



PATIENT CASE STUDY REPORT

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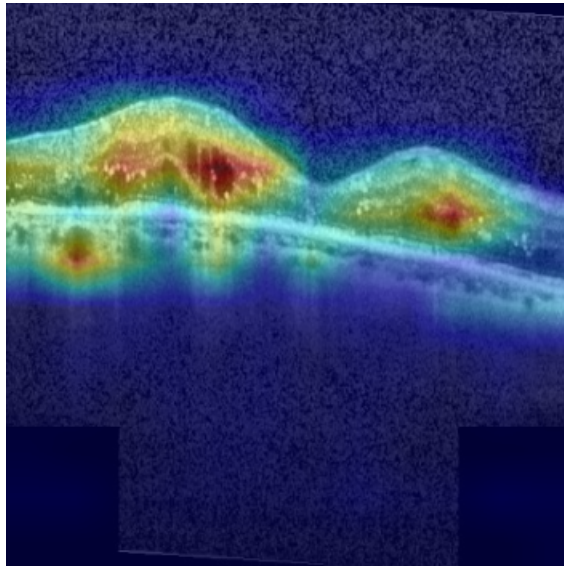
[H1 - Gatekeeper Triage]
Status : Abnormal (100.0%)
Decision : Pass to H2

[H2 - Pathology Routing]
True : DME
Top-1 : DME (96.2%)
Top-2 : CNV (3.6%)
Status : [CORRECT]

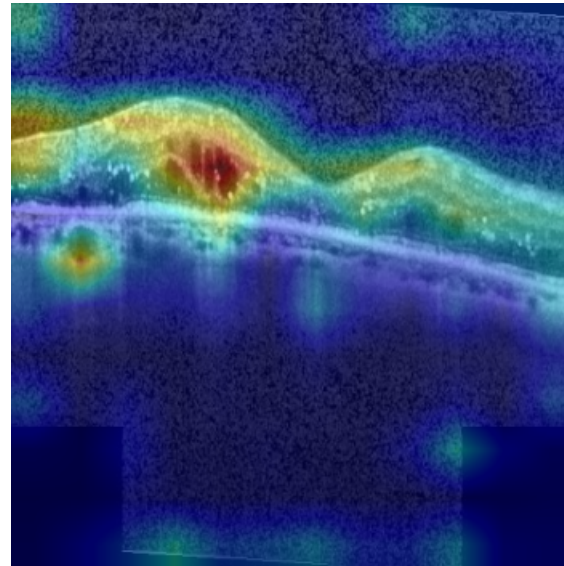
[CAM Ambiguity Quantification]
Cosine Sim : 0.830
Overlap IoU : 0.320

Pathology Probabilities (>5%):
- DME : 96.2%

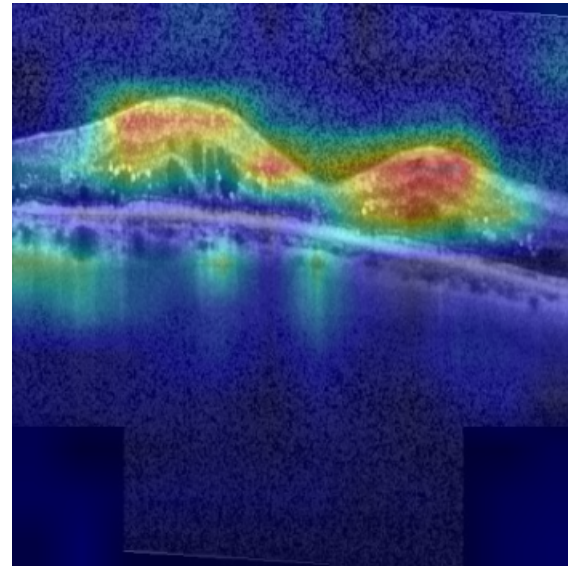
H1 Grad-CAM
(Triage)



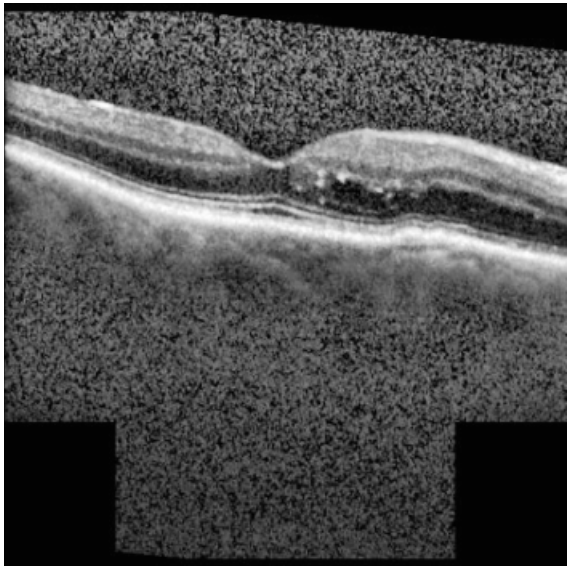
H2 Grad-CAM
DME (96.2%)



H2 Grad-CAM
CNV (3.6%)



Original OCT Scan
True: DME



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PATIENT CASE STUDY REPORT
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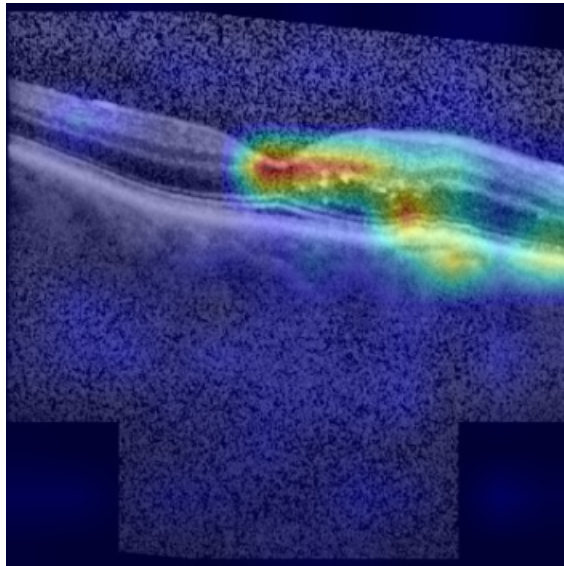
[ H1 - Gatekeeper Triage ]
Status   : Abnormal (97.5%)
Decision : Pass to H2

[ H2 - Pathology Routing ]
True     : DME
Top-1    : CNV (59.5%)
Top-2    : DRUSEN (26.5%)
Status   : [MISMATCH]

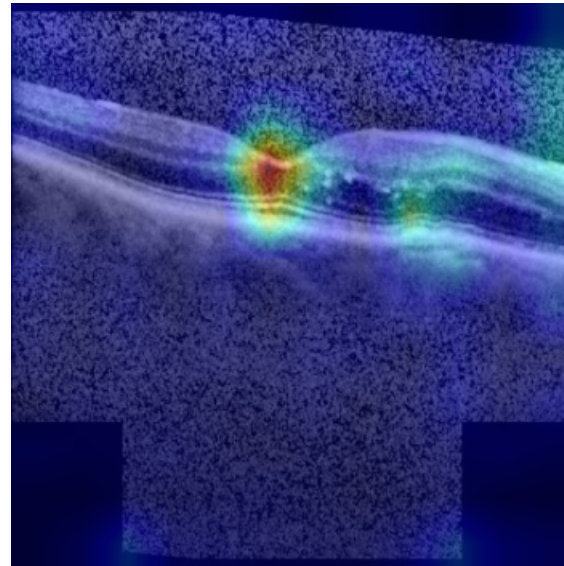
[ CAM Ambiguity Quantification ]
Cosine Sim : 0.662
Overlap IoU : 0.010

Pathology Probabilities (>5%):
- CNV       : 59.5%
- DRUSEN    : 26.5%
- DME       : 10.7%
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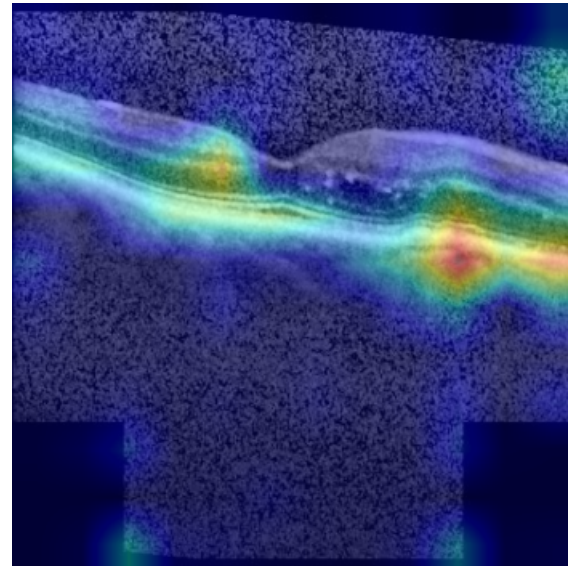
H1 Grad-CAM
(Triage)



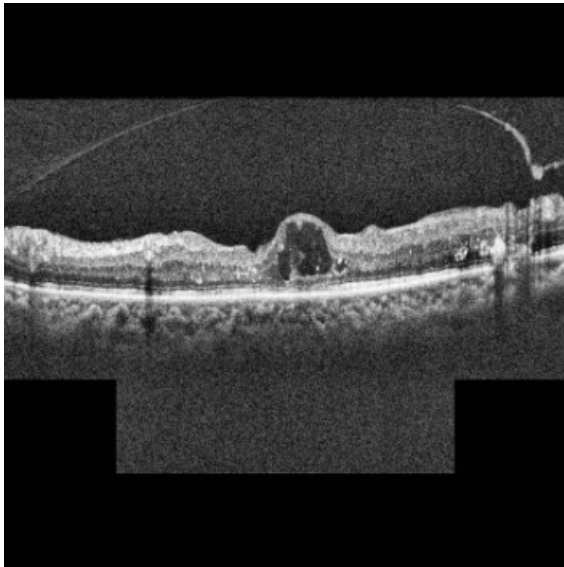
H2 Grad-CAM
CNV (59.5%)



H2 Grad-CAM
DRUSEN (26.5%)



Original OCT Scan
True: DR



PATIENT CASE STUDY REPORT

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[H1 - Gatekeeper Triage]

Status : Abnormal (100.0%)

Decision : Pass to H2

[H2 - Pathology Routing]

True : DR

Top-1 : DR (92.8%)

Top-2 : DME (6.9%)

Status : [CORRECT]

[CAM Ambiguity Quantification]

Cosine Sim : 0.518

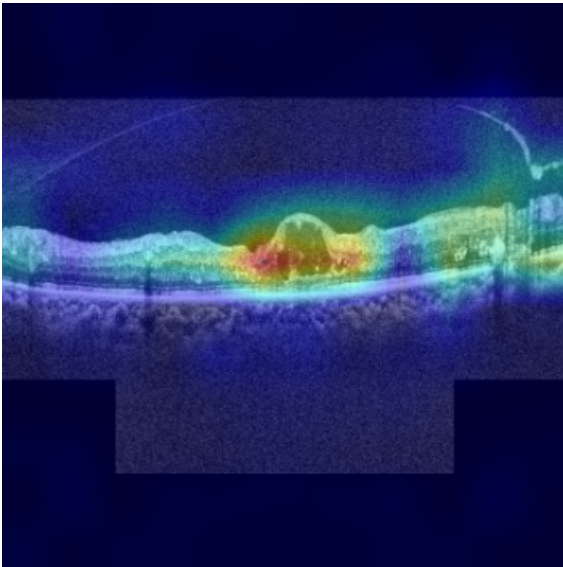
Overlap IoU : 0.000

Pathology Probabilities (>5%):

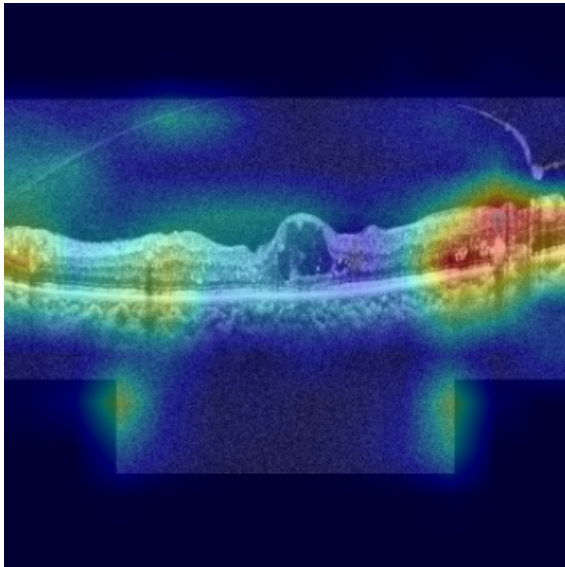
- DR : 92.8%

- DME : 6.9%

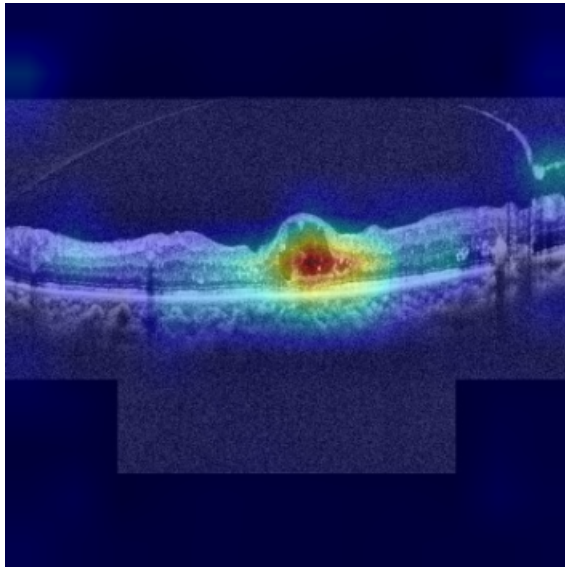
H1 Grad-CAM
(Triage)



H2 Grad-CAM
DR (92.8%)



H2 Grad-CAM
DME (6.9%)



Original OCT Scan
True: DR



PATIENT CASE STUDY REPORT
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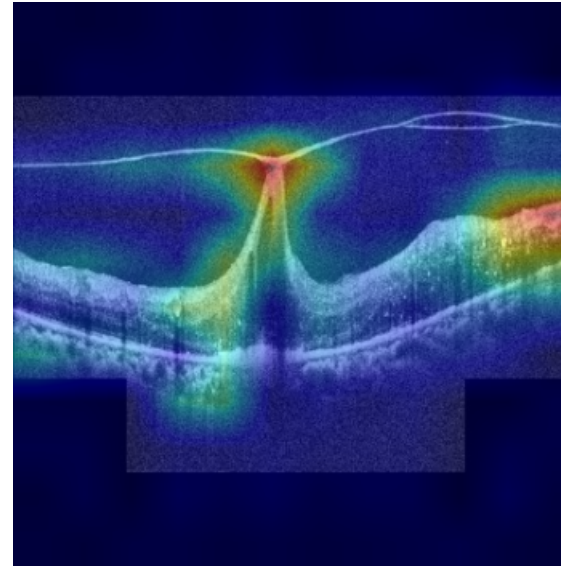
[H1 - Gatekeeper Triage]
Status : Abnormal (100.0%)
Decision : Pass to H2

[H2 - Pathology Routing]
True : DR
Top-1 : DR (98.8%)
Top-2 : DME (1.0%)
Status : [CORRECT]

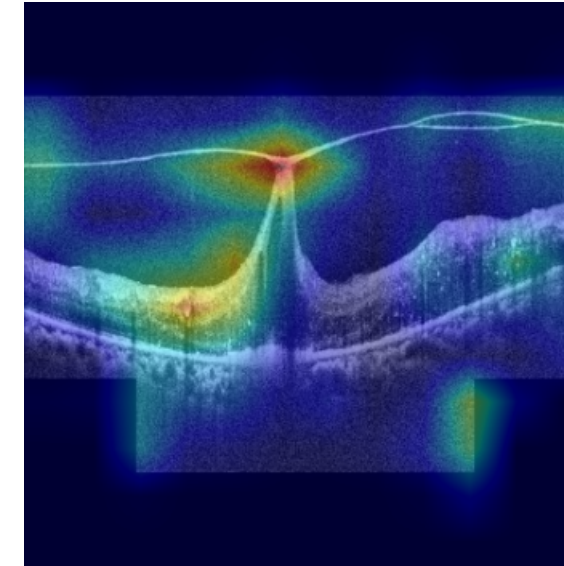
[CAM Ambiguity Quantification]
Cosine Sim : 0.603
Overlap IoU : 0.000

Pathology Probabilities (>5%):
- DR : 98.8%

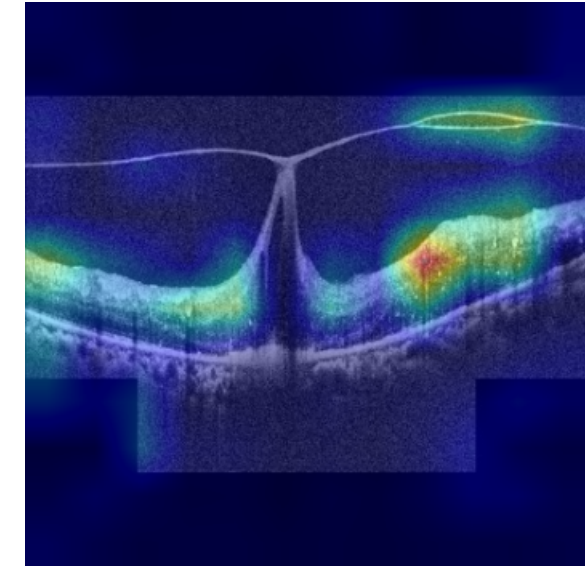
H1 Grad-CAM
(Triage)



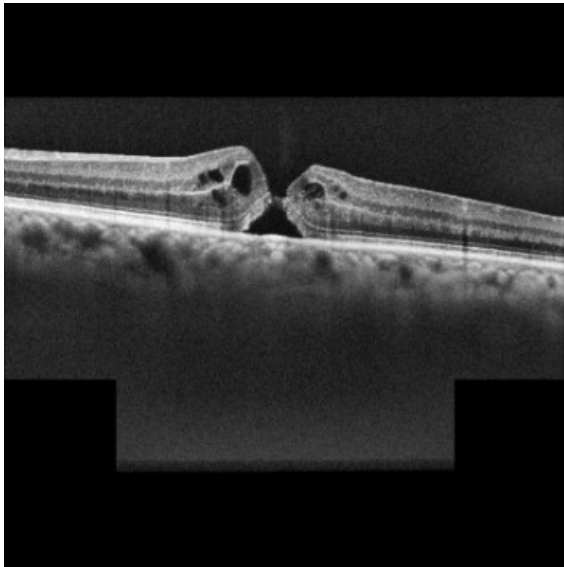
H2 Grad-CAM
DR (98.8%)



H2 Grad-CAM
DME (1.0%)



Original OCT Scan
True: MH



PATIENT CASE STUDY REPORT
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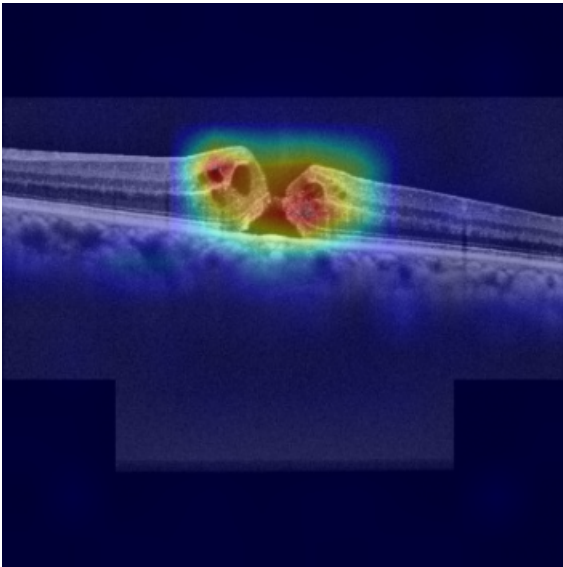
[H1 - Gatekeeper Triage]
Status : Abnormal (100.0%)
Decision : Pass to H2

[H2 - Pathology Routing]
True : MH
Top-1 : MH (99.0%)
Top-2 : DME (0.9%)
Status : [CORRECT]

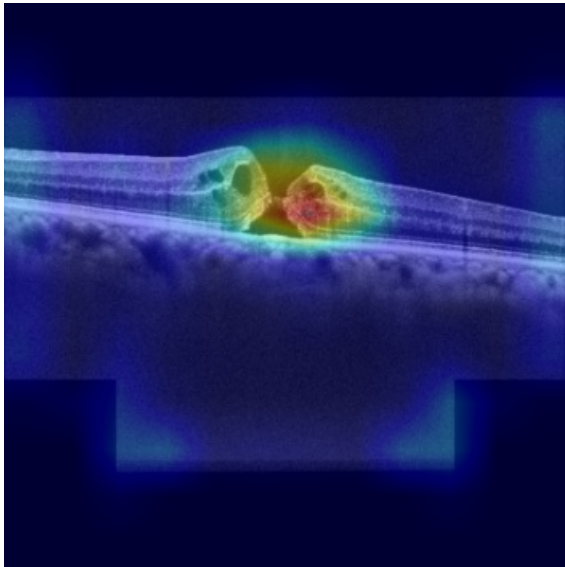
[CAM Ambiguity Quantification]
Cosine Sim : 0.598
Overlap IoU : 0.062

Pathology Probabilities (>5%):
- MH : 99.0%

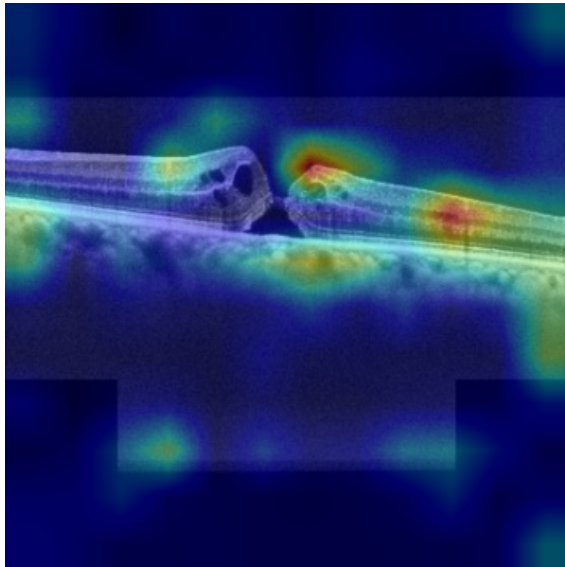
H1 Grad-CAM
(Triage)



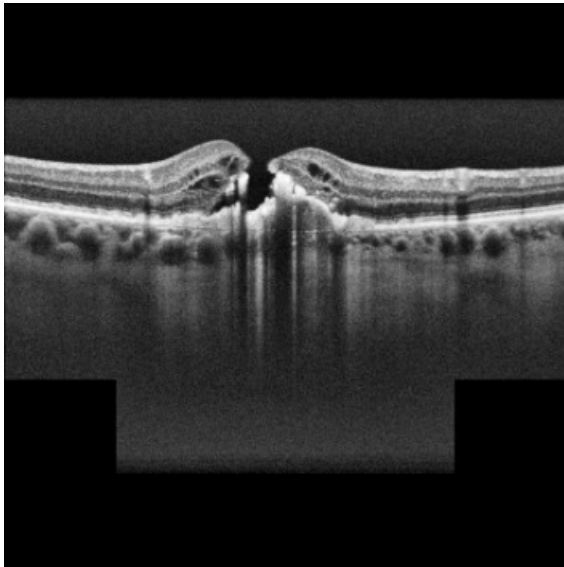
H2 Grad-CAM
MH (99.0%)



H2 Grad-CAM
DME (0.9%)



Original OCT Scan
True: MH



PATIENT CASE STUDY REPORT

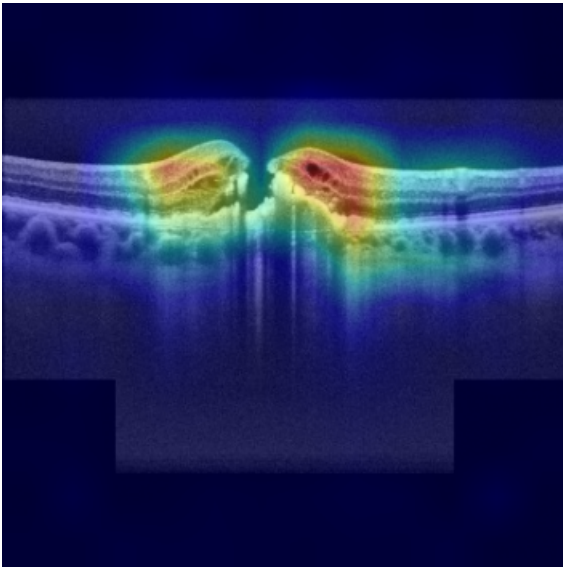
[H1 - Gatekeeper Triage]
Status : Abnormal (100.0%)
Decision : Pass to H2

[H2 - Pathology Routing]
True : MH
Top-1 : CNV (56.3%)
Top-2 : MH (13.7%)
Status : [MISMATCH]

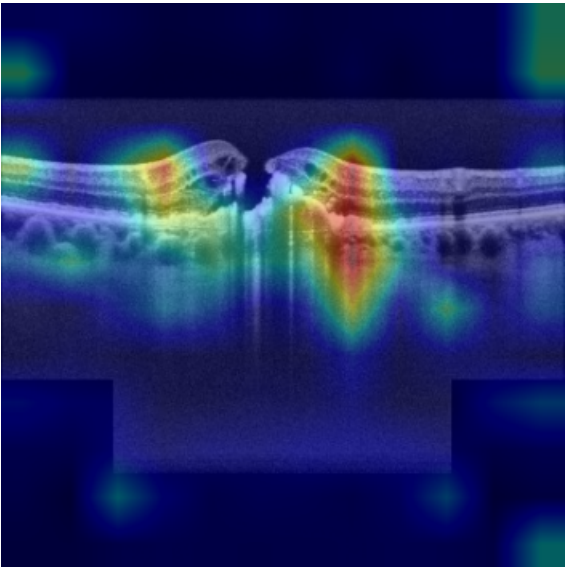
[CAM Ambiguity Quantification]
Cosine Sim : 0.293
Overlap IoU : 0.009

Pathology Probabilities (>5%):
- CNV : 56.3%
- MH : 13.7%
- DME : 13.4%
- General_AMD : 11.9%

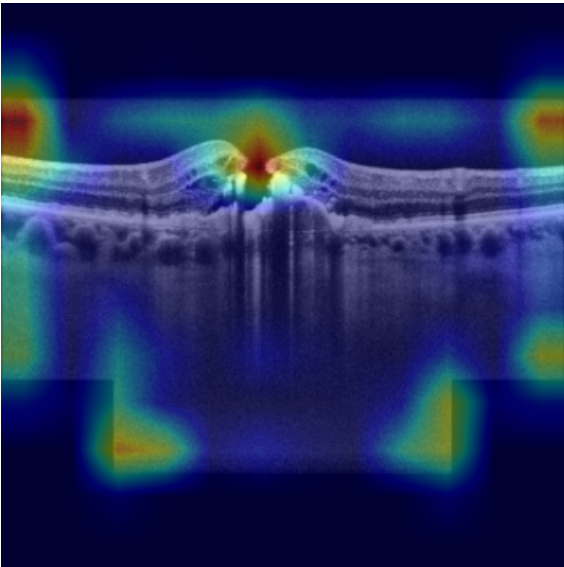
H1 Grad-CAM
(Triage)



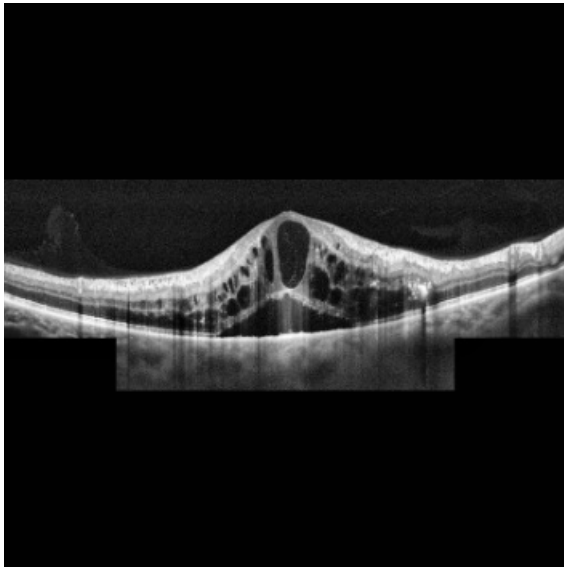
H2 Grad-CAM
CNV (56.3%)



H2 Grad-CAM
MH (13.7%)



Original OCT Scan
True: RVO



```
PATIENT CASE STUDY REPORT
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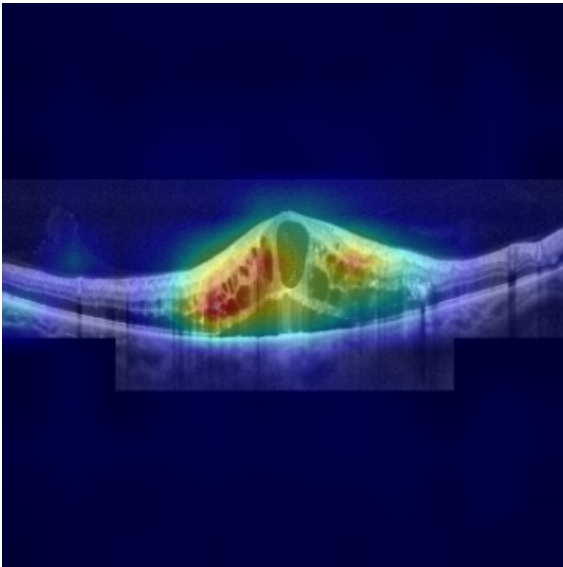
[ H1 - Gatekeeper Triage ]
Status   : Abnormal (100.0%)
Decision : Pass to H2

[ H2 - Pathology Routing ]
True     : RVO
Top-1    : DME (97.8%)
Top-2    : RVO (1.6%)
Status   : [MISMATCH]

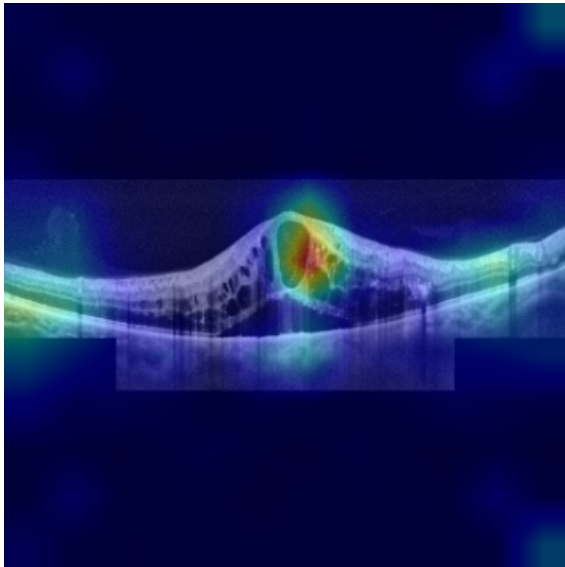
[ CAM Ambiguity Quantification ]
Cosine Sim : 0.514
Overlap IoU : 0.053

Pathology Probabilities (>5%):
- DME           : 97.8%
```

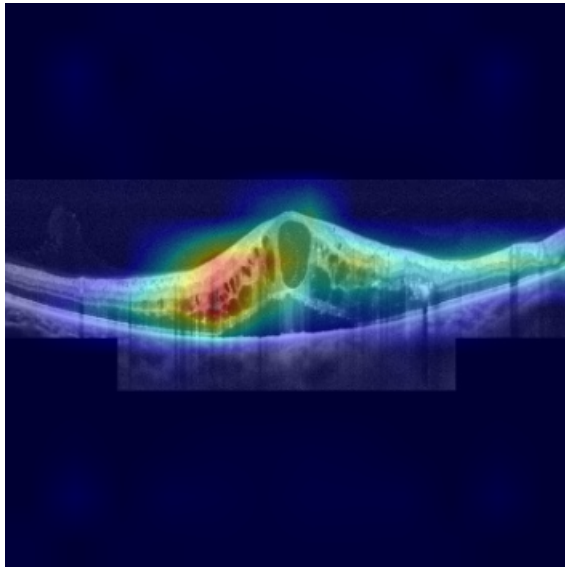
H1 Grad-CAM
(Triage)



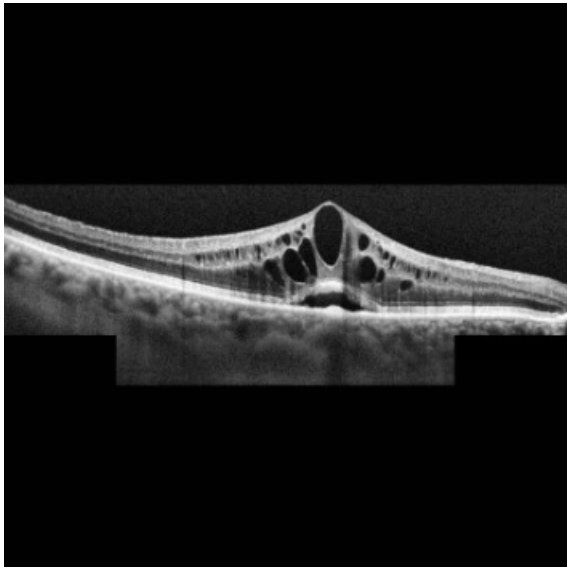
H2 Grad-CAM
DME (97.8%)



H2 Grad-CAM
RVO (1.6%)



Original OCT Scan
True: RVO



PATIENT CASE STUDY REPORT

=====

[H1 - Gatekeeper Triage]
Status : Abnormal (100.0%)
Decision : Pass to H2

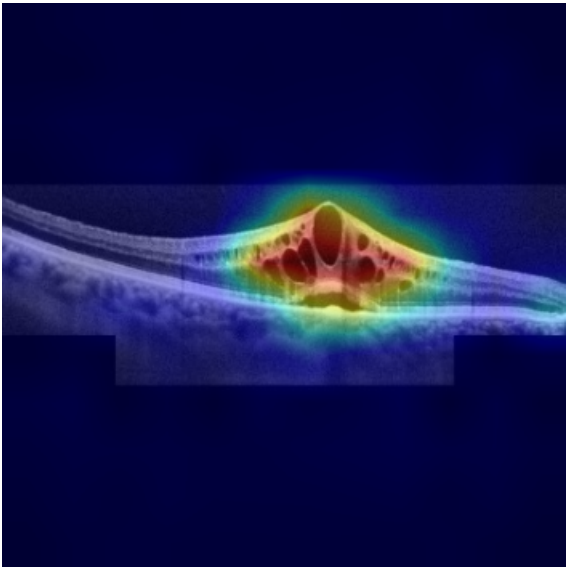
[H2 - Pathology Routing]
True : RVO
Top-1 : MH (52.1%)
Top-2 : DME (47.6%)
Status : [MISMATCH]

[CAM Ambiguity Quantification]
Cosine Sim : 0.632
Overlap IoU : 0.000

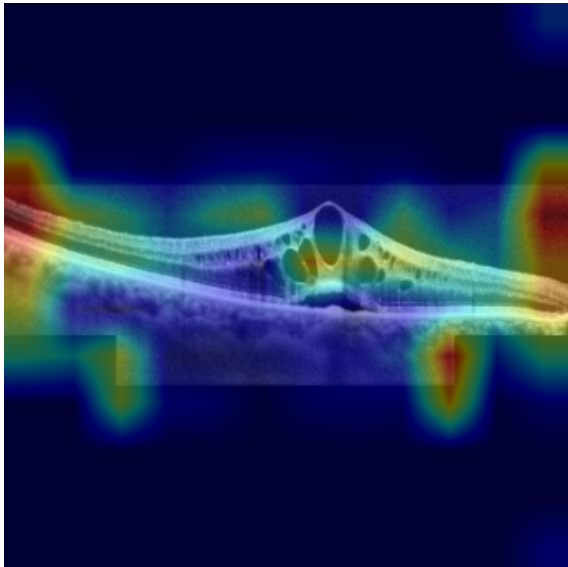
Pathology Probabilities (>5%):

- MH : 52.1%
- DME : 47.6%

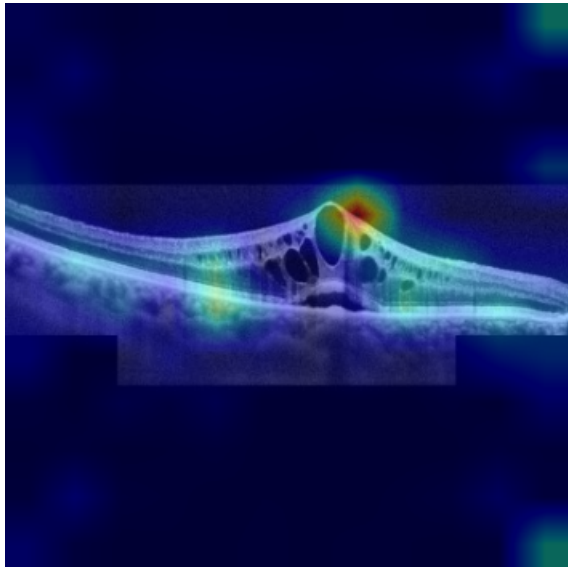
H1 Grad-CAM
(Triage)



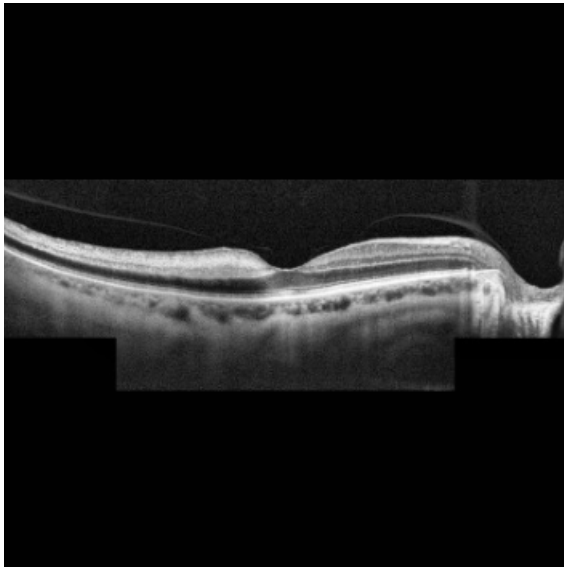
H2 Grad-CAM
MH (52.1%)



H2 Grad-CAM
DME (47.6%)



Original OCT Scan
True: RAO



PATIENT CASE STUDY REPORT

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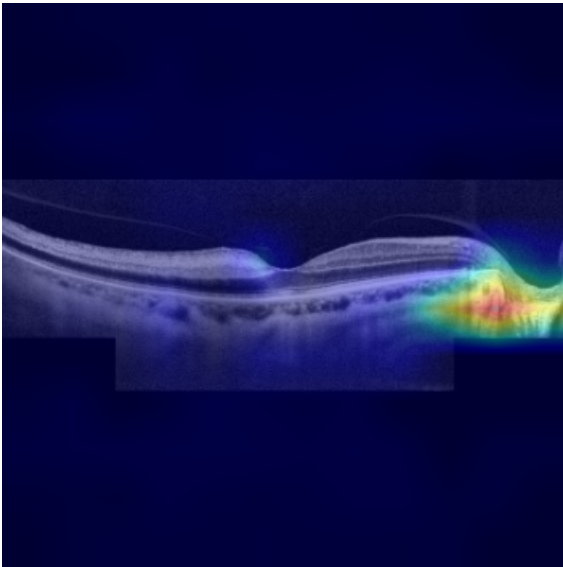
[H1 - Gatekeeper Triage]
Status : Abnormal (72.0%)
Decision : Pass to H2

[H2 - Pathology Routing]
True : RAO
Top-1 : RAO (58.0%)
Top-2 : MH (41.2%)
Status : [CORRECT]

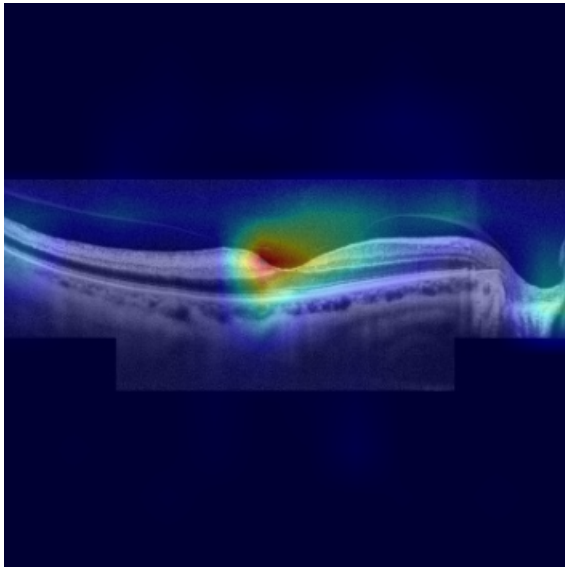
[CAM Ambiguity Quantification]
Cosine Sim : 0.402
Overlap IoU : 0.000

Pathology Probabilities (>5%):
- RAO : 58.0%
- MH : 41.2%

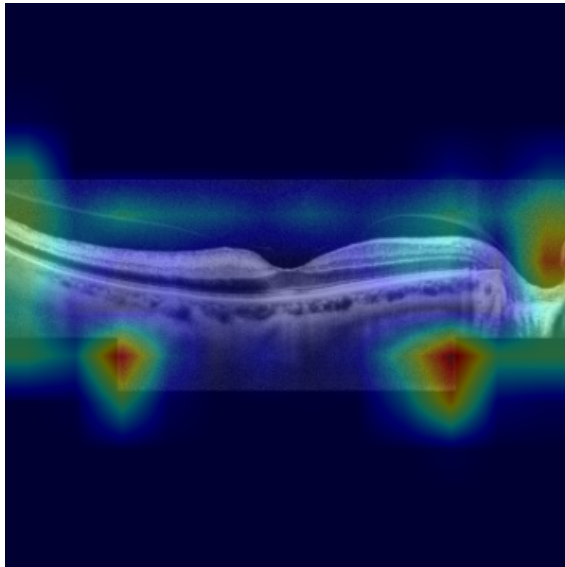
H1 Grad-CAM
(Triage)



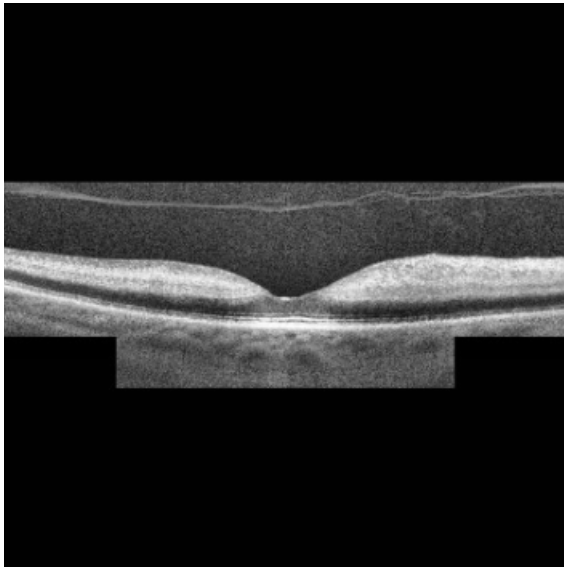
H2 Grad-CAM
RAO (58.0%)



H2 Grad-CAM
MH (41.2%)



Original OCT Scan
True: RAO



PATIENT CASE STUDY REPORT
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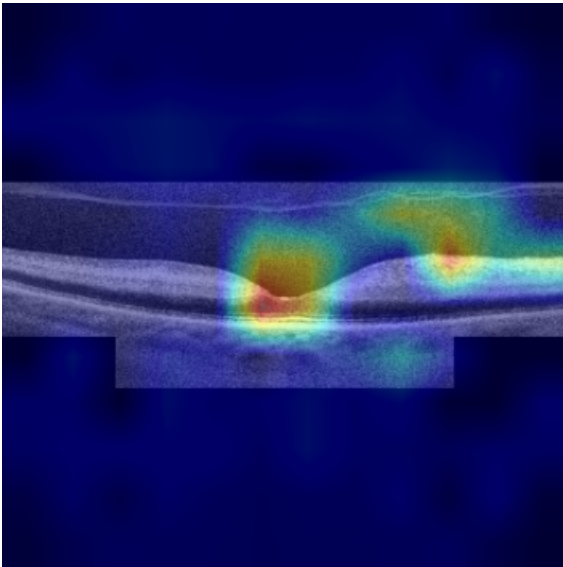
[H1 - Gatekeeper Triage]
Status : Normal (47.6%)
Decision : Normal Scan

[H2 - Pathology Routing]
True : RAO
Top-1 : RAO (100.0%)
Top-2 : CSR (0.0%)
Status : [CORRECT]

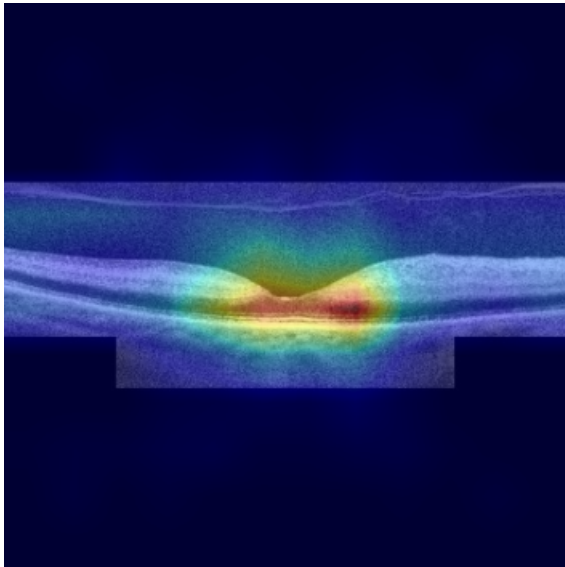
[CAM Ambiguity Quantification]
Cosine Sim : 0.379
Overlap IoU : 0.000

Pathology Probabilities (>5%):
- RAO : 100.0%

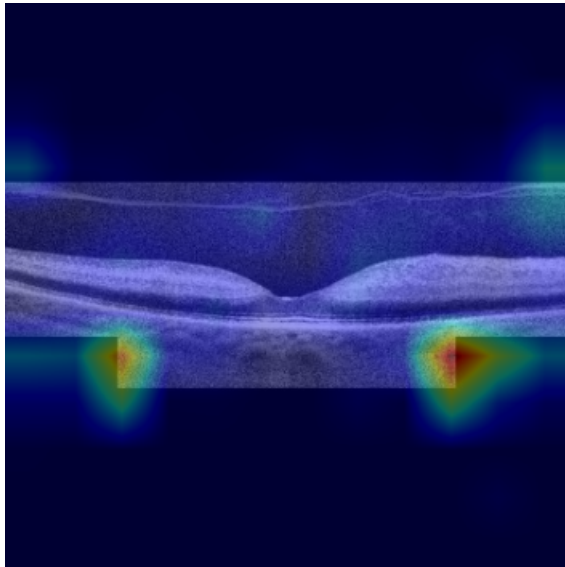
H1 Grad-CAM
(Triage)



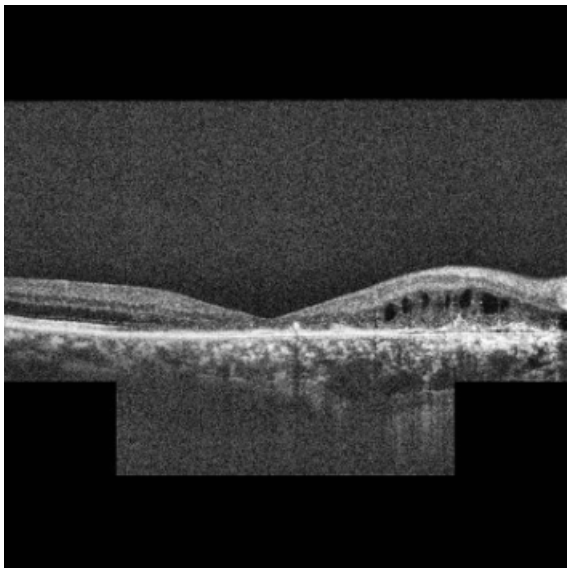
H2 Grad-CAM
RAO (100.0%)



H2 Grad-CAM
CSR (0.0%)



Original OCT Scan
True: CSR



PATIENT CASE STUDY REPORT

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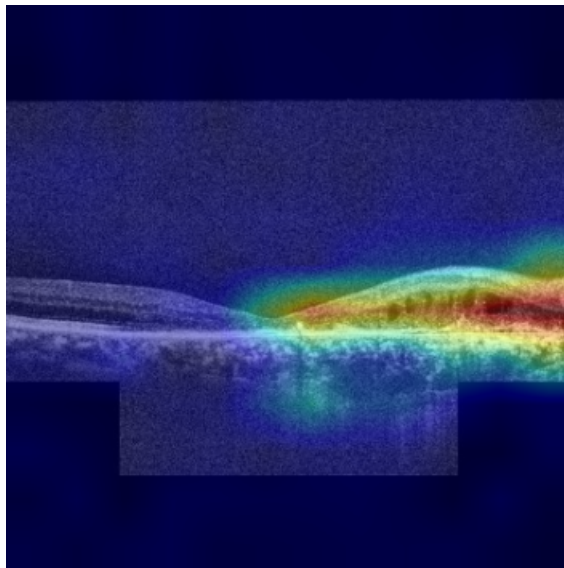
[H1 - Gatekeeper Triage]
Status : Abnormal (100.0%)
Decision : Pass to H2

[H2 - Pathology Routing]
True : CSR
Top-1 : CSR (77.1%)
Top-2 : DR (12.6%)
Status : [CORRECT]

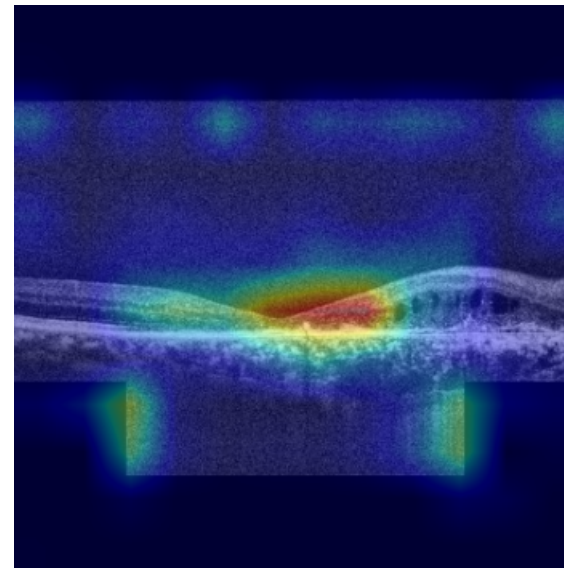
[CAM Ambiguity Quantification]
Cosine Sim : 0.541
Overlap IoU : 0.004

Pathology Probabilities (>5%):
- CSR : 77.1%
- DR : 12.6%
- General_AMD : 6.5%

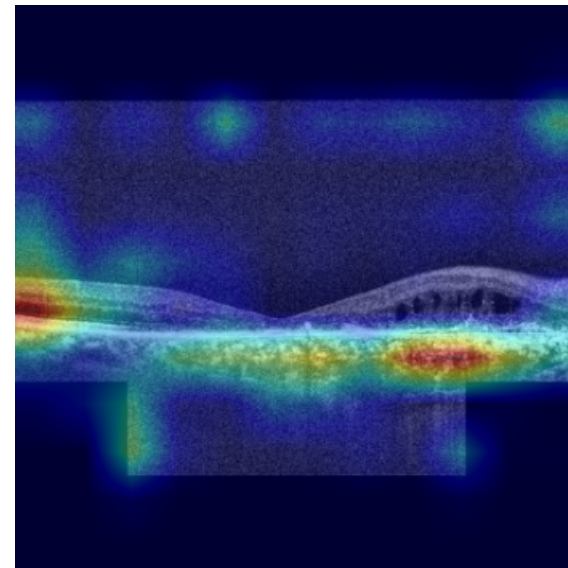
H1 Grad-CAM
(Triage)



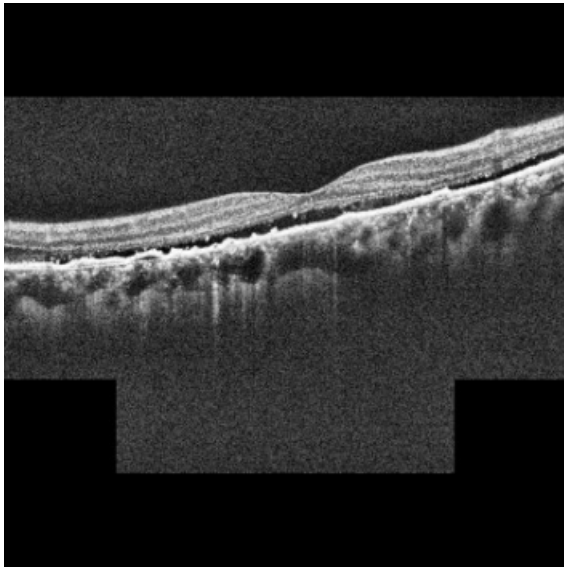
H2 Grad-CAM
CSR (77.1%)



H2 Grad-CAM
DR (12.6%)



Original OCT Scan
True: CSR



PATIENT CASE STUDY REPORT
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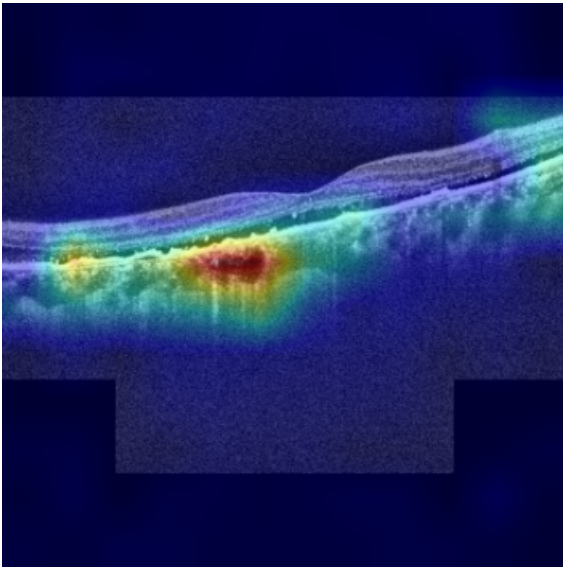
[H1 - Gatekeeper Triage]
Status : Abnormal (100.0%)
Decision : Pass to H2

[H2 - Pathology Routing]
True : CSR
Top-1 : CSR (99.4%)
Top-2 : General_AMD (0.5%)
Status : [CORRECT]

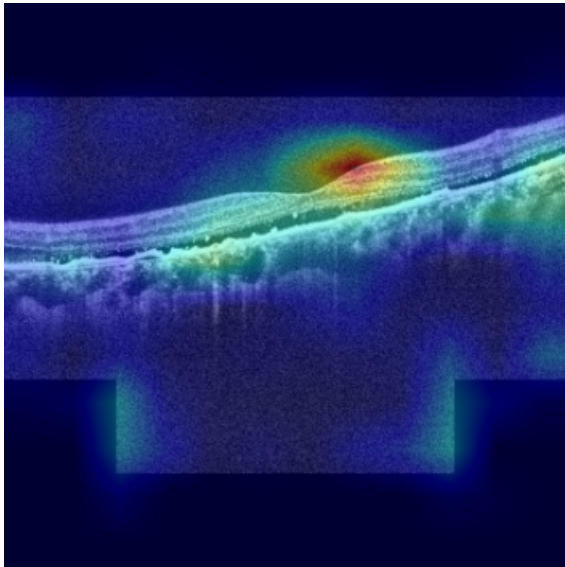
[CAM Ambiguity Quantification]
Cosine Sim : 0.794
Overlap IoU : 0.089

Pathology Probabilities (>5%):
- CSR : 99.4%

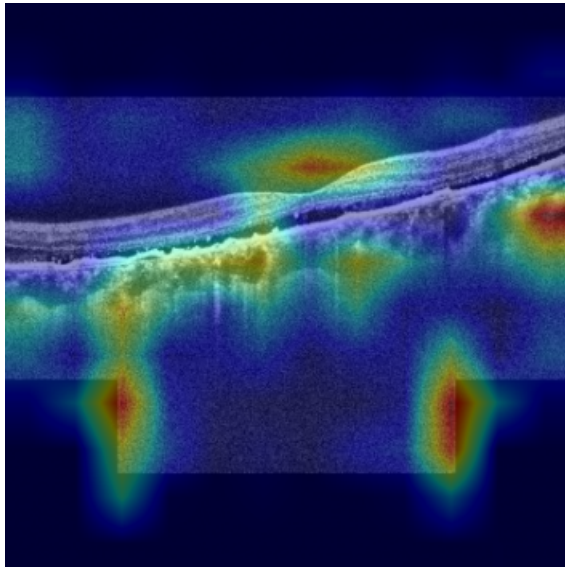
H1 Grad-CAM
(Triage)



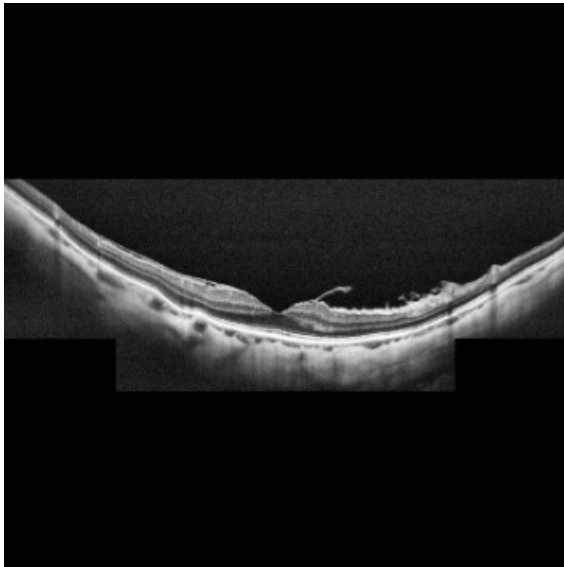
H2 Grad-CAM
CSR (99.4%)



H2 Grad-CAM
General_AMD (0.5%)



Original OCT Scan
True: ERM



PATIENT CASE STUDY REPORT
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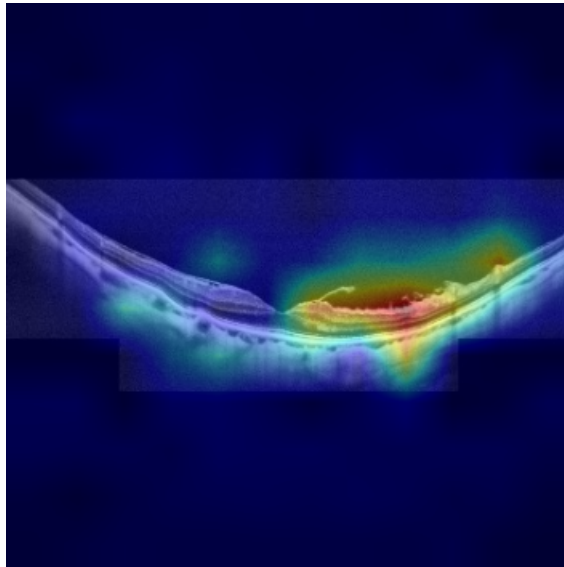
[H1 - Gatekeeper Triage]
Status : Abnormal (100.0%)
Decision : Pass to H2

[H2 - Pathology Routing]
True : ERM
Top-1 : ERM (83.3%)
Top-2 : DME (8.1%)
Status : [CORRECT]

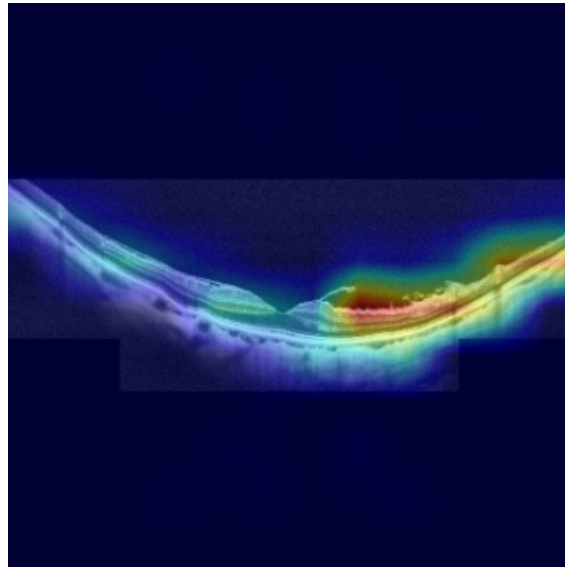
[CAM Ambiguity Quantification]
Cosine Sim : 0.422
Overlap IoU : 0.068

Pathology Probabilities (>5%):
- ERM : 83.3%
- DME : 8.1%
- VID : 5.5%

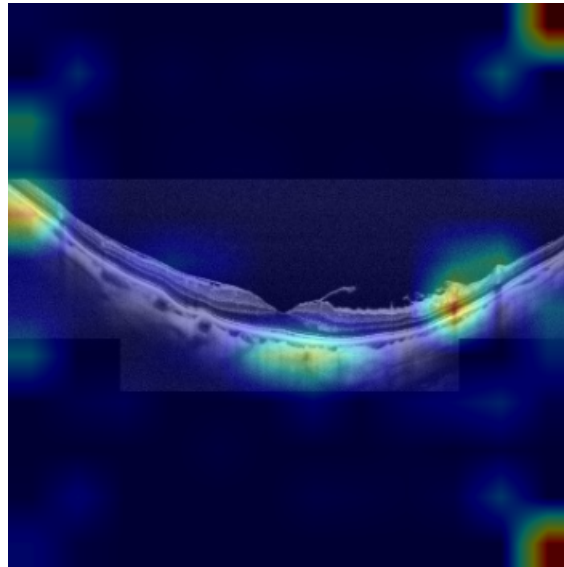
H1 Grad-CAM
(Triage)



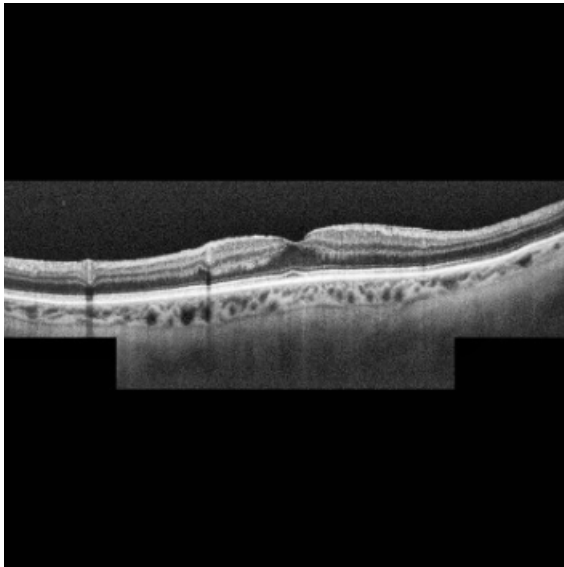
H2 Grad-CAM
ERM (83.3%)



H2 Grad-CAM
DME (8.1%)



Original OCT Scan
True: ERM



PATIENT CASE STUDY REPORT
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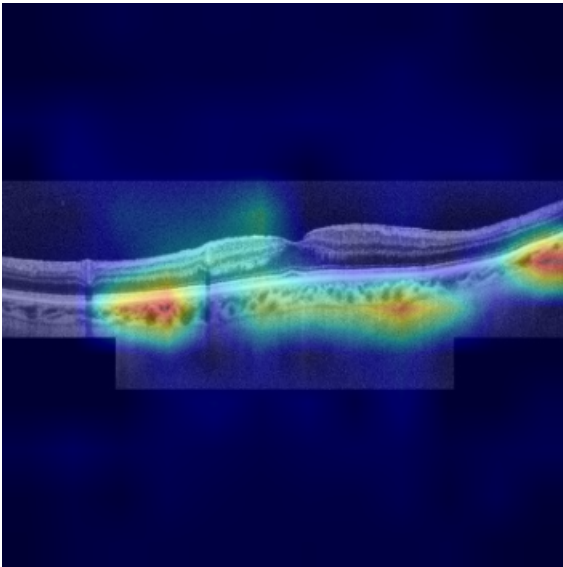
[H1 - Gatekeeper Triage]
Status : Abnormal (99.2%)
Decision : Pass to H2

[H2 - Pathology Routing]
True : ERM
Top-1 : ERM (96.8%)
Top-2 : DME (2.5%)
Status : [CORRECT]

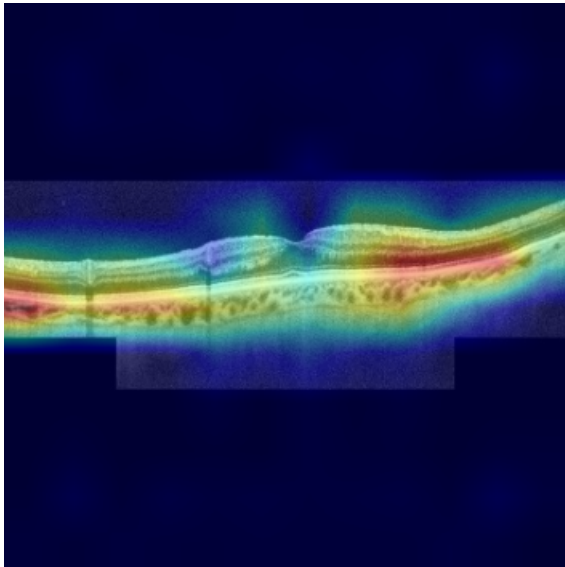
[CAM Ambiguity Quantification]
Cosine Sim : 0.593
Overlap IoU : 0.135

Pathology Probabilities (>5%):
- ERM : 96.8%

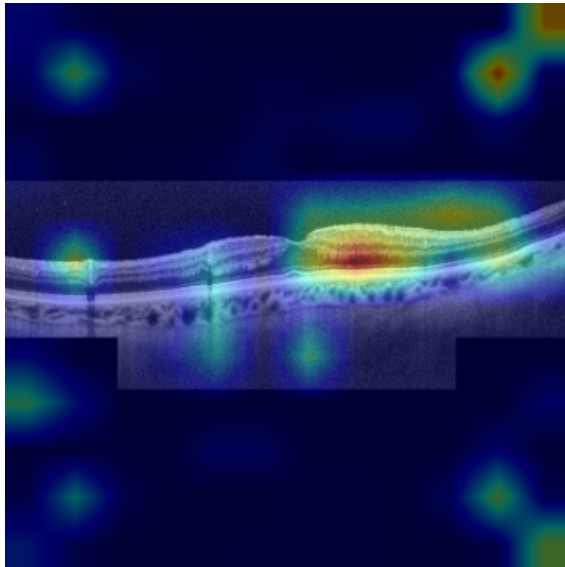
H1 Grad-CAM
(Triage)



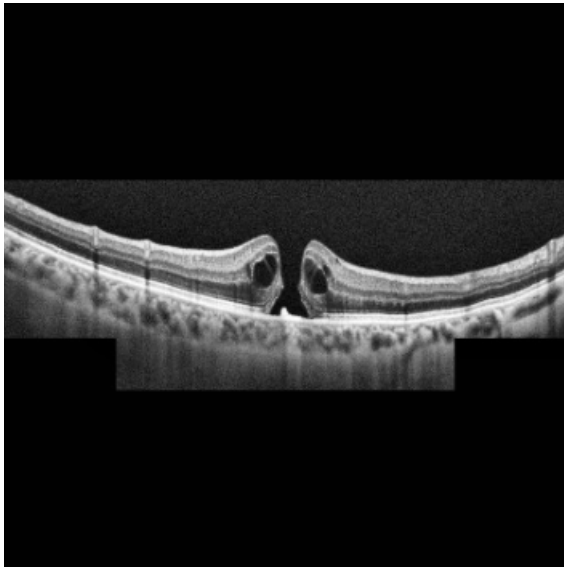
H2 Grad-CAM
ERM (96.8%)



H2 Grad-CAM
DME (2.5%)



Original OCT Scan
True: VID



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PATIENT CASE STUDY REPORT
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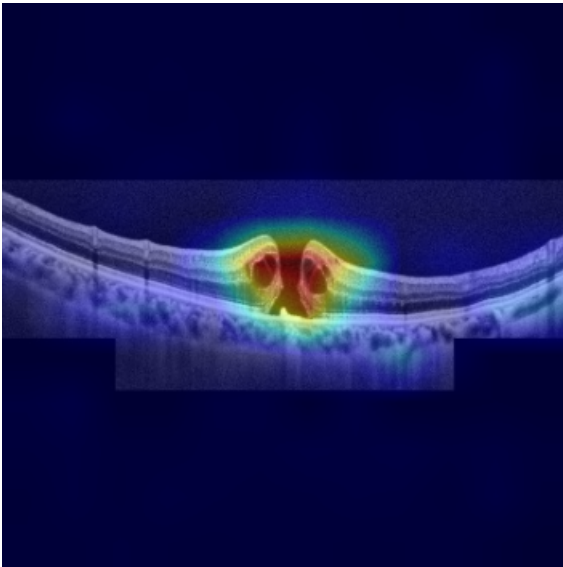
[ H1 - Gatekeeper Triage ]
Status   : Abnormal (100.0%)
Decision : Pass to H2

[ H2 - Pathology Routing ]
True     : VID
Top-1    : MH (65.9%)
Top-2    : DME (27.2%)
Status   : [MISMATCH]

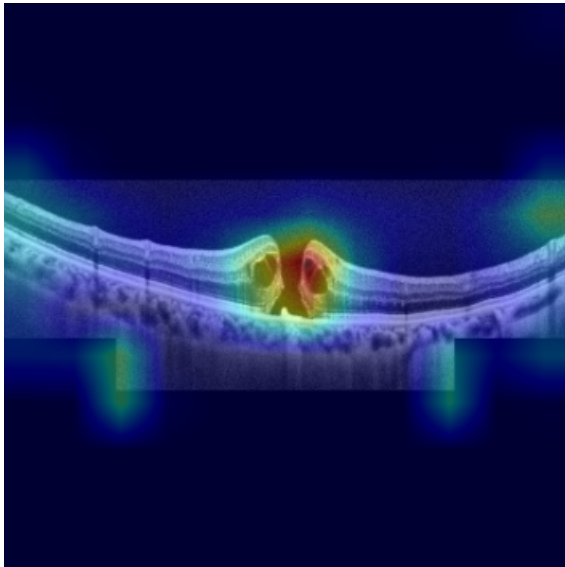
[ CAM Ambiguity Quantification ]
Cosine Sim : 0.593
Overlap IoU : 0.029

Pathology Probabilities (>5%):
- MH       : 65.9%
- DME      : 27.2%
- VID      : 5.1%
```

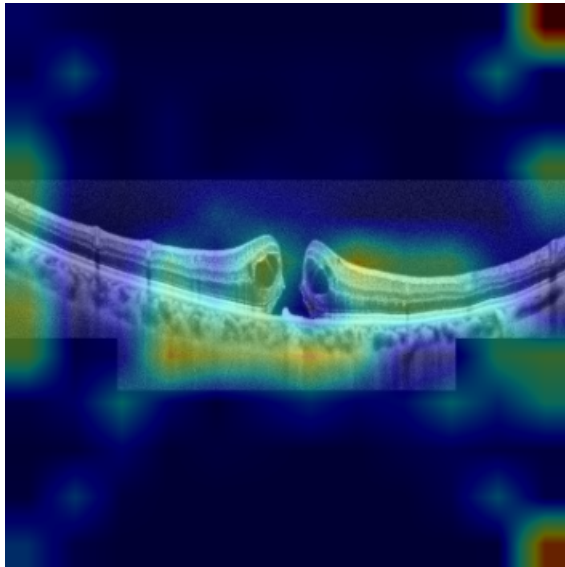
H1 Grad-CAM
(Triage)



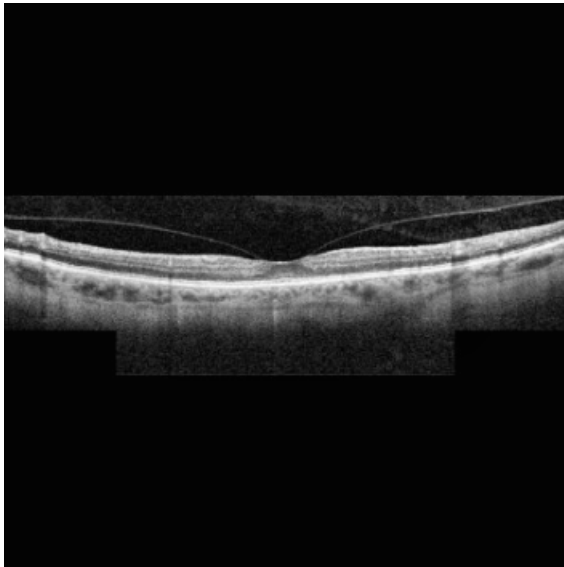
H2 Grad-CAM
MH (65.9%)



H2 Grad-CAM
DME (27.2%)



Original OCT Scan
True: VID



PATIENT CASE STUDY REPORT

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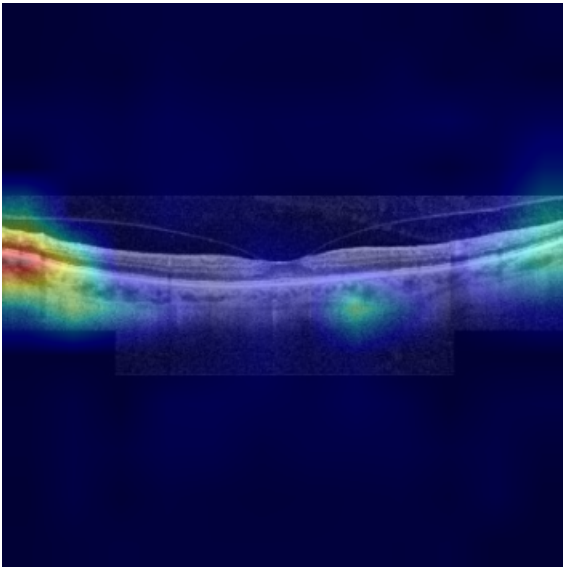
[H1 - Gatekeeper Triage]
Status : Abnormal (99.8%)
Decision : Pass to H2

[H2 - Pathology Routing]
True : VID
Top-1 : MH (56.4%)
Top-2 : VID (19.6%)
Status : [MISMATCH]

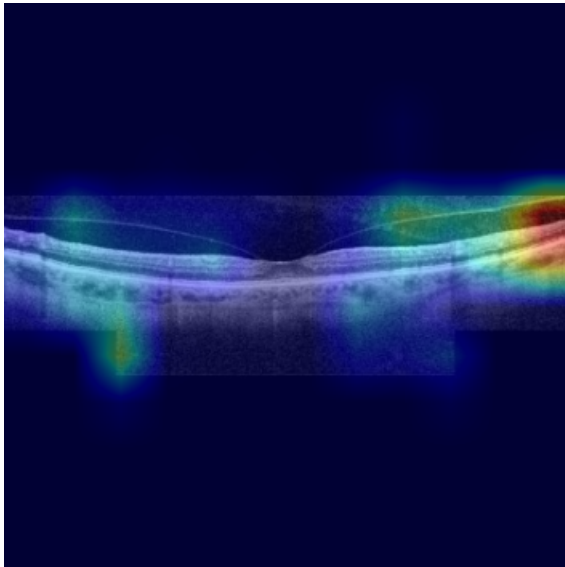
[CAM Ambiguity Quantification]
Cosine Sim : 0.788
Overlap IoU : 0.300

Pathology Probabilities (>5%):
- MH : 56.4%
- VID : 19.6%
- DME : 18.7%

H1 Grad-CAM
(Triage)



H2 Grad-CAM
MH (56.4%)



H2 Grad-CAM
VID (19.6%)

